

# Newcomers Help Most After Being Helped, Veterans Do Not: Evidence from Stack Overflow

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## Abstract

Whether people who receive help online go on to help others is an open question. Prior observational studies report mixed results, and designs that ignore prior user activity may overstate reciprocity. We develop a matched difference-in-differences survival analysis of over 21 million Stack Overflow questions, comparing users whose questions were answered with propensity-score-matched users whose questions were not. Rather than asking whether reciprocity works in general, we locate the narrow conditions under which it moves behavior. Users whose questions were answered helped others slightly more often (increment at answer arrival, hazard ratio  $[HR] = 1.049$ , 95% CI  $[1.044, 1.054]$ ). The response depended on tenure: newcomers responded most ( $HR = 1.071$  in the first week, 95% CI  $[1.056, 1.086]$ ), while the longest-tenured users did not respond measurably. The tenure gradient is the most robust pattern here. Timing mattered too, though far more fragilely: users answered within 30 to 60 minutes responded most strongly, while answers under 30 minutes show no elevation. Most of our data predate 2020, and the newcomer response does not survive a split at ChatGPT’s release, so we read these estimates as a pre-generative-AI baseline rather than a general test of reciprocity. Gratitude, moral obligation, social expectation, and incentive displacement remain proposed mechanisms rather than measured ones. On this one platform, generalized reciprocity works less as a general engine of cooperation than as a contributor-recruitment channel at the point of entry: designers gain most by getting newcomers’ questions answered.

*Keywords:* generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

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## 1. Introduction

Whether people who receive help on an online platform go on to help others has been a persistent question in the study of online communities.

Generalized reciprocity, the tendency to help others after being helped oneself (Nowak and Roch, 2007; Baker and Bulkley, 2014), is among the mechanisms proposed to sustain cooperation on platforms like Stack Overflow and Wikipedia, where the same two people rarely exchange help directly. If people who receive help reliably go on to give it, cooperation may sustain itself through cascading helping acts, even among strangers (Fowler and Christakis, 2010). The behavioral evidence, however, is inconsistent: some observational studies find substantial reciprocity effects (Baker and Bulkley, 2014; Chen et al., 2019), while others find weak or null relationships (Wasko and Faraj, 2005; Joyce and Kraut, 2006). We argue that much of this inconsistency is methodological. Our main contribution is a cleaner test on one platform: whether users help more shortly after someone answers their own question. After laying out how different behavioral mechanisms that may lead to reciprocity may play out on the platform we study, we use our design to test where, for whom, and under what timing generalized reciprocity occurs. While we cannot directly measure the psychological mechanisms that theory proposes, namely gratitude, moral obligation, social expectation, and incentive displacement, our design cleanly tests the behavioral consequence those mechanisms predict.

Researchers who study reciprocity on platforms face a stubborn confound: more active users are generally more likely to post questions and answers. Studies that correlate help received with help given over an aggregate period of time, without accounting for the temporal order of events, conflate reciprocity with general user activity (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). Prior designs have also largely treated receiving help as a one-time event, even though people ask for and give help continuously.

We address both problems by developing a matched difference-in-differences (DiD) survival analysis, applied to over 21 million questions from Stack Overflow. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users whose question received an answer (treated) and propensity-score-matched users whose question did not (control). We use Cox proportional hazards models, which accommodate the continuous timing of helping and the staggered arrival of answers across users.

We develop two theoretical predictions alongside the basic reciprocity hypothesis (H1). Drawing on theories of socialization and norm internalization, we predict that reciprocity is stronger among newcomers than among expe-

rienced users (H2). Newcomers have not yet internalized community-specific norms or learned to value platform incentives such as reputation points; general moral obligations to reciprocate are therefore more likely to drive their behavior. As users accumulate experience, these general-purpose triggers may be displaced by community-specific motivations. We also predict that faster responses amplify the reciprocity effect (H3), because users are more grateful for quick answers while their need is still present (McCullough et al., 2001; Tsang, 2006).

The results support the baseline reciprocity hypothesis (H1) in a modest form, and also corroborate the moderation by tenure (H2) and answer speed (H3). Compared with their matched controls and with their own pre-question baseline, users who receive an answer subsequently answer other people’s questions more often: the increase in propensity at answer arrival is  $HR = 1.049$  (95% CI [1.044, 1.054],  $p < .001$ ). How strongly users respond depends on how long they have been on the platform: newcomers respond most ( $HR = 1.07$  in the first week,  $p < .001$ ), and the longest-tenured users do not respond measurably at all. While fast answers generally lead to a stronger reciprocity effect — the moderation of the arrival increment by log response time is negative in every tenure bucket, and significant in all but the longest-tenured ( $\gamma$  between  $-0.02$  and  $-0.09$ ; Table 8) — we find that users who wait between 30 and 60 minutes show the strongest reciprocity ( $HR = 1.19$ , 95% CI [1.17, 1.21],  $p < .001$ ; Table 9). We test and discuss possible explanations further below.

Our study makes three contributions. First, using a design built to remove the activity confound that inflates observational estimates, we provide a cleaner short-term test: users who receive an answer help more afterward ( $HR = 1.05$ ), but by much less than prior observational studies report. Second, reciprocity is strongest among newcomers, and tenured users show none. The moderation by tenure is the most robust pattern in the data. Third, we find an inverted U-shaped relationship in which a moderate response time carries the largest effect, and develop different explanations with implications for platform design based on this.

Two boundaries follow from the design. The first is that the effect we identify is narrow. It is concentrated among users in their first week, it peaks for answers arriving between 30 and 60 minutes, and it is small in absolute terms even where it is strongest. The second is that the estimates are dated. Most questions stem from before 2020, and the newcomer effect does not survive a split at the release of ChatGPT. We therefore read our

results as a pre-generative-AI baseline for one platform rather than as a general test of whether reciprocity sustains cooperation. Nonetheless, our design can flexibly be applied to new data from other platforms. We discuss implications for the theory of generalized reciprocity, platform designers, and future platform research in the age of generative AI.

## 2. Theory

### 2.1. Generalized Reciprocity

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*, the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (Nowak and Roch, 2007; Fowler and Christakis, 2010). Unlike direct reciprocity, it requires no repeated exchange with the same person: it operates as a pay-it-forward principle, sustaining collective contributions in groups — such as online knowledge-sharing platforms — where the same two people rarely interact twice.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional response. On the moral side, individuals reason by universalization, asking what would happen if all members requested more help than they gave. Widespread free-riding would undermine the community as a source of help, and this conclusion generates a shared obligation to reciprocate toward the group (Levine et al., 2020; Curry et al., 2019). On the emotional side, gratitude aroused by the perceived generosity of others motivates prosocial action beyond direct reciprocation (McCullough et al., 2001; Tsang, 2006; Bartlett and DeSteno, 2006) and is especially relevant where free riding is easy, as on knowledge-sharing platforms (Molm et al., 2007; Elster, 2009). These moral and emotional pathways are the *proposed* generative mechanisms behind generalized reciprocity.

A third pathway runs through visibility rather than through gratitude or obligation. Users might feel observed when they receive help, and helping afterwards may be a response to being observed while being helped as a matter of *social expectation*. This pathway could be described as reputation-induced generalized reciprocity, since it works through perceived social expectation rather than morality or gratitude through the help a user received (Simpson et al., 2018; Milinski et al., 2002). We carry it alongside the other two as a

proposed mechanism, and return in Section 6.5 to what our design can and cannot say about it.

Evidence that generalized reciprocity can operate and can initiate cascading acts of generosity (Fowler and Christakis, 2010; Boyd and Richerson, 1989) is robust: comparative research indicates that the underlying norm is globally shared (Curry et al., 2019), and experimental demonstrations span laboratory (Greiner and Levati, 2005; Molm et al., 2007; Stanca, 2009) and field settings (Mujcic and Leibbrandt, 2018; Fowler and Christakis, 2010). The open question is how strongly it operates among real users on large platforms — a question observational data have answered inconsistently, for reasons we argue are largely methodological.

## *2.2. Empirical Challenges in Measuring Generalized Reciprocity*

One challenge is that much of the evidence relies on what contributors say about their own motives, in surveys and interviews. Contributors frequently describe reciprocity as a central reason for their engagement (Wasko and Faraj, 2000; Lakhani, 2003). Yet people have difficulty articulating the moral motives behind their actions (Vaisey, 2009; McClelland et al., 1989), and introspection struggles to separate reciprocity from altruism, conformity to perceived norms, status pursuit, or the popular rhetoric of “giving back” as post-hoc rationalization (Batson, 1991; Bénabou and Tirole, 2006). Self-reported data may therefore overstate the prevalence of reciprocity as a discrete motive.

The deeper challenge is that the people who ask for help are also the people most likely to give it. Users who ask questions on platforms like Stack Overflow answer them at disproportionately high rates. Not because receiving help triggered a reciprocal response, but because both behaviors reflect generally high user activity on the platform. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). This confound is the central identification challenge that motivates the matched DiD design we develop in Section 4.

Prior observational studies of online reciprocity report mixed magnitudes. Baker and Bulkley (2014) and Chen et al. (2019) find positive associations between help received and subsequent helping; Wasko and Faraj (2005) and Joyce and Kraut (2006) find weak or null relationships. Our matched DiD estimates sit below the larger positive reports, consistent with the view that

designs that do not net out baseline activity overstate the effect. Our finding of clear generalized reciprocity among newcomers, which decreases to null among the most tenured users, offers one reconciliation of the mixed prior findings.

### *2.3. Reciprocity and the User Trajectory*

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has not received proper attention: the user trajectory from newcomer to experienced contributor. We argue that generalized reciprocity is strongest for new users and attenuates as users gain experience on the platform.

This prediction draws on theories of socialization and norm internalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives, such as reputation points, badges, or leaderboard positions, carry little motivational weight because the newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (Levine et al., 2020; Curry et al., 2019).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement echoes accounts of how external incentives can crowd out moral motives (Bénabou and Tirole, 2006).

Two further mechanisms accelerate this decline. As users accumulate experiences of receiving help, each additional instance becomes less surprising, and the gratitude that may fuel reciprocity fades with repetition (McCullough et al., 2008). And as users learn that the platform rewards helping

with reputation and status, they may increasingly attribute others' helpfulness to status pursuit rather than genuine generosity, further reducing their gratitude.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. It also helps explain why prior studies have reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). *Receiving help increases a user's propensity to help others (generalized reciprocity effect).*

HYPOTHESIS 2 (H2). *The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.*

The stakes of this prediction extend beyond reconciling mixed findings. If the reciprocity association is strongest among newcomers, generalized reciprocity would be less as a general driver of cooperation than as a *contributor-recruitment* channel at the point of entry, before community-specific incentives take hold. We develop the design implication of this finding in Section 6.4.

#### *2.4. Response Time and the Strength of Reciprocity*

The emotional mechanisms underlying reciprocity suggest that the speed of help matters for its strength. Gratitude, the primary proposed emotional driver of generalized reciprocity (McCullough et al., 2001), is triggered by the perception of a benefactor's generosity, and its intensity is likely to depend on the characteristics of the helping experience including its timeliness.

A swift response can amplify the experience of receiving help: a fast answer signals that the community is attentive and generous, and it arrives while the asker's need is still salient. In that case, gratitude should be strongest. Conversely, when an answer arrives hours or days later, the original need may have faded or been met elsewhere, the cognitive link between asking and receiving weakens, and the delay itself may signal an unresponsive community. Each of these changes attenuates the emotional and moral responses that underpin reciprocation.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). *Faster responses to a user’s question strengthen the generalized reciprocity effect.*

This hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize how quickly help arrives and how salient the helping experience is. Those accounts imply that reciprocity grows as help arrives sooner. Platform settings, however, introduce session dynamics, i.e. whether the asker is still online when the answer arrives, that lab experiments and social psychological theories abstract away. We therefore treat the prediction as directional, test whether it holds monotonically, and return to this qualification when presenting the results.

### 3. Empirical Setting: Stack Overflow

Stack Overflow is well suited to examining generalized reciprocity in online knowledge sharing. The platform publishes quarterly data dumps that record what every user did and when, reaching back to its inception. From these records, we can reconstruct how people helped one another over time, and in what order they received and gave help. This time series of events is what our study of generalized reciprocity builds on. Stack Overflow’s active question-and-answer volume has declined sharply since the release of generative AI tools in late 2022 (del Rio-Chanona et al., 2024; Burtch et al., 2024); our sample spans platform inception through 1 April 2025 and therefore covers a period that mostly predates, but extends into, the steep decline marked by the release of ChatGPT. Most of our data records the platform while it was proliferating. We treat the findings as a historical snapshot of one large, technical, relatively anonymous, status-driven platform; external validity to other communities is discussed in Section 6.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of June 2025, shortly after our sample cutoff, the site had accumulated over 24 million questions and 36 million answers from more than 29 million registered users (Stack Exchange, 2025). Its user base spans a wide range of experience levels, from learners to professional software developers (Stack Overflow, 2024). Questions on the platform must be tightly focused on specific coding and software problems; broad or opinion-based inquiries are typically closed by community moderation (Stack Exchange Inc., 2024). The most frequently discussed topics, as reflected in question tags, include JavaScript, Python, Java, C#, PHP, Android, and

HTML, spanning areas from web development and mobile applications to data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; and reputation points are earned for valued contributions such as providing high-quality answers (Stack Exchange Inc., 2024).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers. This creates a timestamped sequence (question posted, answer received) that we exploit to measure the temporal dynamics of generalized reciprocity. On Stack Overflow, question-askers and answerers are typically different individuals, which makes the platform well suited to studying *generalized* reciprocity: users who receive help from one community member may subsequently help entirely different members.

## 4. Data and Methods

To test our predictions, we develop a matched staggered DiD survival analysis that exploits the order in which users ask for and give help on Stack Overflow. Our design compares changes in helping behavior (measured as the hazard of posting an answer) before and after a user receives an answer, between users who receive an answer and matched users who do not. This approach combines the staggered DiD framework of Callaway and Sant’Anna (2021), which accommodates variation in treatment timing and treatment effect heterogeneity, with the counting-process formulation of the Cox proportional hazards model (Andersen and Gill, 1982), which allows treatment status and other covariates to vary continuously over the observation window.

### 4.1. Study Design

For each question posted by a user, we construct a timeline centered on two key events: the posting of the question ( $T_{\text{question}}$ ) and, for questions that receive an answer, the time at which the first answer with a non-negative vote score is posted ( $T_{\text{answer}}$ ). The observation window opens two days before the question ( $T_{\text{question}} - 2D$ ) and closes two days *after the answer arrives* ( $T_{\text{answer}} + 2D$ ). The window is therefore anchored on the answer rather than on the question: the pre-question and post-answer periods each have a fixed two-day duration, while the intervening waiting period runs for the question’s response time. The total observation window is  $4D + RT$ , with longer response

times resulting in longer windows. This gives each treated question a full two-day post-answer period in which to observe changes in helping behavior.

We define three phases within each observation window (see Figure 1). The first phase, the *pre-question phase*, is from the start of the observation window to the posting of the question ( $T_{\text{question}} - 2D$  to  $T_{\text{question}}$ ). This phase captures baseline helping activity before the focal question is asked. The second phase, the *waiting period*, is from the posting of the question to the receipt of the first answer ( $T_{\text{question}}$  to  $T_{\text{answer}}$ ). During this interval, the user has asked for help but has not yet received it. The final *post-answer phase* is from the receipt of the answer to the end of the observation window ( $T_{\text{answer}}$  to  $T_{\text{answer}} + 2D$ ), a fixed two-day span regardless of how long the answer took to arrive. This is the phase in which the reciprocity effect, if it exists, should manifest.

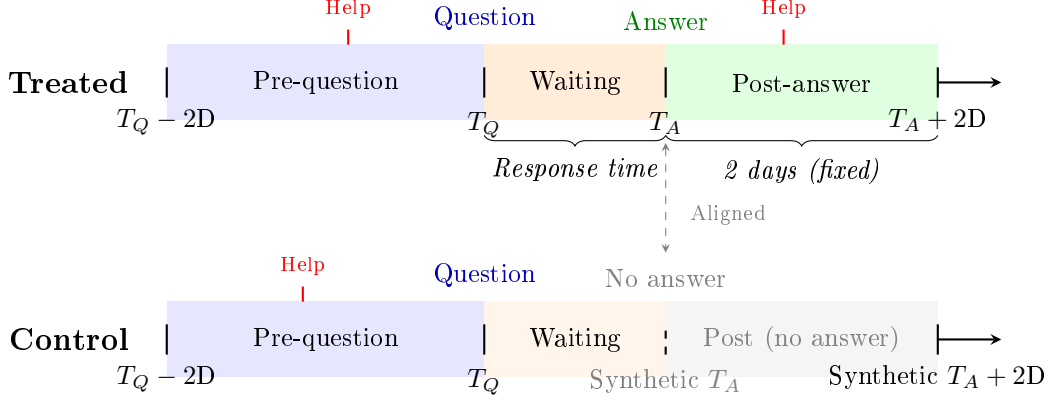
A key feature of our design is that the post-answer (treated) phase begins at a different elapsed time for each observation, depending on how quickly the question receives a response.

This staggered timing of the post-answer phase corresponds to the setting studied by Callaway and Sant’Anna (2021), who show that standard two-way fixed-effects estimators can be biased when treatment timing varies, and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying change in treatment status within each observation window and by comparing treated observations only to not-yet-treated or never-treated controls.

Control users (those whose questions did not receive an answer) have no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer was posted 4 hours after their question, the matched control observation’s timeline is split at  $T_{\text{question}} + 4h$  and its window closes at  $T_{\text{question}} + 4h + 2D$ , creating an aligned “post” phase of the same two-day duration. Because the control inherits the treated response time, matched treated and control observations share identical window boundaries and identical two-day post-answer durations, so any differences between them in the post-answer phase are not driven by differences in the length of the observation period.

We center the survival clock in our Cox models on the posting of the question: time zero corresponds to the start of the observation window ( $T_{\text{question}} - 2D$ ), so the question itself lands at day 2 of the timeline. Because the window

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



*Note.* The observation window begins two days before the question is posted ( $T_Q$ ) and ends two days after the answer arrives ( $T_A + 2D$ ). The pre-question and post-answer periods therefore each last two days, while the waiting period between  $T_Q$  and  $T_A$  varies with response time. The total window length is  $4D + RT$ . The treated timeline (top) contains three periods: the pre-question period, the waiting period, and the post-answer period following the first qualifying answer. The control timeline (bottom) represents a propensity-score-matched question without an answer. Its post-question period is split at a synthetic transition point equal to the treated user’s response time to align the two timelines. Variation in  $T_A$  across observations motivates the staggered difference-in-difference design.

closes two days after the answer, its end falls at day 4 + response time rather than at a fixed day 4. A question answered within the first day therefore closes near day 4, while a question answered after three days closes near day 7. We record every help event (an answer posted by the focal user) at its exact timestamp and convert it to elapsed time on this clock.

These helping events constitute the outcome in our survival analysis. The decomposition (Table 6) also examines broader definitions that count comments and edits. Conceptually, our design asks: Does the hazard of helping increase after a user receives an answer, relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the descriptive help rates over this timeline in Section 5 (Figure 3), before turning to the formal Cox model estimates.

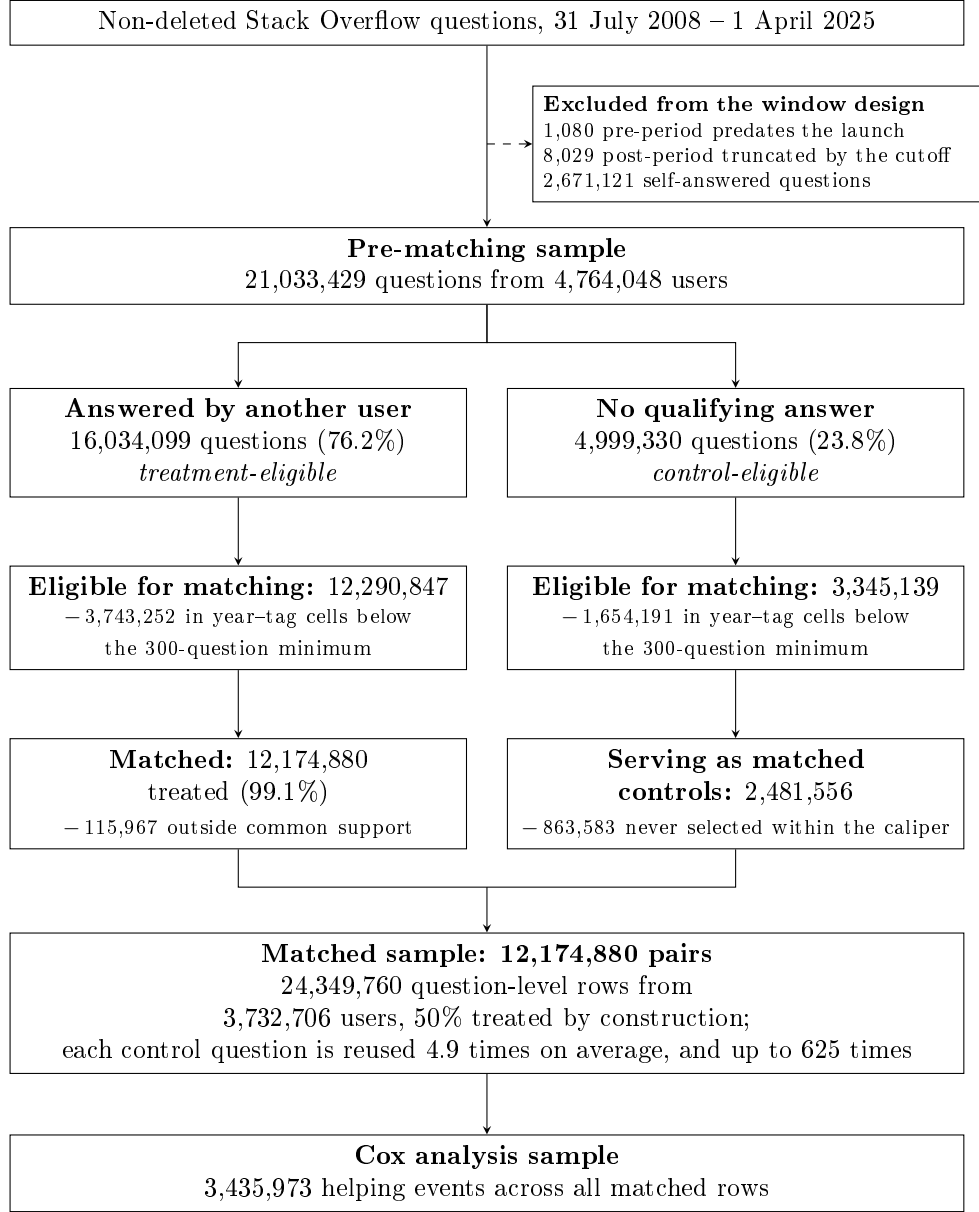
#### 4.2. Data

We used the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We dropped all 9,109 ques-

tions where our observation window was truncated either by the platform launch date or the cutoff date. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users, of which 16,034,099 (76.2%) received at least one answer from another user (treatment-eligible) and 4,999,330 (23.8%) received no answer (control-eligible). Matching (described below) pairs questions only within cells defined by calendar year and primary tag, so we drop cells holding fewer than 300 questions, as these are too thin to support a within-cell comparison. This leaves 12,290,847 answered and 3,345,139 unanswered questions (15,635,986 in total) eligible for matching. Of the 12,290,847 eligible treated questions, 12,174,880 (99.1%) were matched to a control; the remaining 115,967 fell outside common support. This attrition is not random: the treated propensity-score density piles up near one (Figure B.6), so the discarded questions are disproportionately those that were almost certain to be answered. The 12,174,880 matched treated questions draw on 2,481,556 distinct control questions, each reused about 4.9 times on average (up to 625 times) to fill the control slots. The matched file therefore comprises 24,349,760 question-level rows (12,174,880 matched pairs) from 3,732,706 unique users, with a 50% treatment rate by construction. All matched rows enter the Cox analysis; the helping outcome is the posting of an answer to another user’s question (answers to one’s own question do not count), yielding 3,435,973 helping events in the pooled analysis sample (Table 5).

Figure 2 traces these steps from the public dump to the matched analysis sample.

Figure 2: Sample Construction: From the Public Dump to the Matched Analysis Sample



*Note.* Appendix C discusses what the observations without common support implies for the scope of the estimand; Appendix E reconciles the sample sizes reported across the Cox tables.

### 4.3. Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

*Outcome.* The event of interest is posting an answer to another user’s question. Within each window we track when (and whether) the focal user answers any other user’s question. The Cox model then estimates the instantaneous hazard of this helping event as a function of time-varying covariates that capture the phase of the observation timeline and the treatment status. Answering one’s own question, and self-directed acts such as accepting an answer, do not count as helping events.

We treat posting answers as the theoretically central helping act. Generalized reciprocity, as modeled in the experimental literature, is a *costly* prosocial contribution: individuals pass a benefit to an unrelated third party at real expense to themselves (Greiner and Levati, 2005; Stanca, 2009; Mujic and Leibbrandt, 2018). Bearing that cost distinguishes reciprocity from low-cost signaling (Molm et al., 2007). Answering a stranger’s question demands substantive effort and yields a public good for the community. We therefore anchor the outcome on answers and report comment- and edit-inclusive alternatives in the decomposition (Table 6); upvotes are one lower-cost channel our data does not support, because the public dump withholds voter identity. To address the concern that answers alone may understate lower-cost reciprocity, we re-estimate the pooled model with lower-cost contributions (comments only, edits only, and summing all these channels together) and report them in Table 6. Previewing the results, this choice appears reasonable: comments alone carry only a slight elevation, well below the answers increment, so reciprocity manifests the strongest in answers and broadening the outcome decreases the estimate (see Section 6.5).

*Treatment.* The treatment is the receipt of an answer. We measure it with *treatment*, a binary variable set to one when the user’s question received, within seven days of posting, at least one answer from *another* user carrying a non-negative score; the seven-day cap keeps the fixed two-day post-answer

Table 1: Overview of Variables

| Variable                        | Description                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Outcome (Survival Event)</i> |                                                                                                                                                                                                                                                                                                            |
| helpEvent                       | A binary event indicator equal to 1 at times when the user posts an answer to another user’s question within the observation window (answers to one’s own question do not count). Broader definitions adding comments and edits are reported in the outcome decomposition.                                 |
| <i>Time-Varying Covariates</i>  |                                                                                                                                                                                                                                                                                                            |
| PostQuestion                    | Binary; 1 for intervals occurring after the user posts their question ( $t \geq T_{\text{question}}$ ), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).                                                                                                      |
| PostAnswer                      | Binary; 1 for intervals occurring after the user receives their first answer ( $t \geq T_{\text{answer}}$ ), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase. |
| treatedPostQuestion             | Interaction of treatment $\times$ phasePostQuestion; captures differential behavior of treated users in the entire period after the question.                                                                                                                                                              |
| isTreatedActive                 | Interaction of treatment $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any change attributable to the act of asking a question.                                                                                 |
| ResponseTime                    | Time between question and the first answer (excluding down-voted answers) of the treated question. We include interactions with all four variables above.                                                                                                                                                  |

window inside the observable record. Control questions are the complement: those that received no such qualifying answer. This definition pools three cases that share the absence of timely, positively scored help from another user: questions never answered, questions answered within seven days only by downvoted posts, and questions whose first qualifying answer arrived after seven days. We choose receiving an answer as the treatment rather than on Stack Overflow’s “accepted answer” mechanism because accepting an answer requires the user to return to the site, and users who return are plausibly also the users who go on to help.

*Response Time.* For each treated observation, the response-time variable used in the interaction terms is  $\log(1 + \text{RT})$ , where  $\text{RT} = t_{\text{answer}} - t_{\text{question}}$  is the actual response time in hours: the elapsed time from the posting of the question to the arrival of the first answer with a non-negative vote score. The discrete response-time bins reported below (Table 9) are likewise cut on response time, so a bin labelled “1–3 days” contains questions answered 24–72 hours after posting. For numerical stability, both response-time interaction terms ( $\text{isTreatedActive} \times \log(1 + \text{RT})$  and  $\text{treatedPostQuestion} \times \log(1 + \text{RT})$ ) are standardized (z-scored) and then clipped to their 5th–95th percentile range before model fitting. Consequently, the reported interaction coefficients  $\gamma$  and  $\delta$  are expressed per one standard deviation of the respective interaction term rather than per unit of  $\log(1 + \text{RT})$ .

*User Experience.* To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using right-closed intervals:  $[0, 7]$ ,  $(7, 30]$ ,  $(30, 180]$ ,  $(180, 365]$ ,  $(365, 1095]$ ,  $(1095, 2190]$ , and  $(2190, \infty)$  days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket lets us examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form.

#### 4.4. Propensity Score Matching

Receiving an answer (the treatment) is correlated with user characteristics: more experienced users, for instance, might ask higher-quality questions that are more likely to be answered. We therefore employ propensity score

matching (Rosenbaum and Rubin, 1983) to improve the comparability of the treatment and control groups. Importantly, we match *across different users*: we pair each treated question with a control question from a *different* user, chosen for its similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the DiD structure within each pair absorbs residual differences.

We estimate the propensity score using pre-treatment covariates that jointly capture user experience, recent activity, and question characteristics, as listed in Table 2. The covariate *tagAcceptShareAvg* addresses the concern that question complexity may confound the treatment: questions in tags with low accepted-answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly topic areas. Matching on the tag-level accepted-answer rate also matters for the response-time analysis, because it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput. We use nearest-neighbor matching *with replacement* and a caliper of 0.05 on the propensity score, requiring in addition an exact match on the calendar year, the question’s primary tag, and the number of tags. Because the pool of unanswered questions is smaller than the pool of answered questions, some control questions are matched to more than one treated question; the matched file therefore contains more rows than unique question pairs (12.2 million pairs, 24.3 million rows). Each row enters the Cox analysis as an independent observation window; standard errors are discussed in the robustness appendix.

Matching narrows the scope of the estimand, and in a way worth stating plainly. Because the exact strata are calendar year, primary tag, and number of tags, a treated question enters the estimate only if an unanswered question exists in the same stratum. Roughly half the unanswered pool has no on-support match, and the control questions that do survive are drawn disproportionately from more readily answered tags. What we estimate is therefore the effect for answered questions in mainstream, well-trafficked tag areas. It is not the effect for the long tail of niche or rarely answered topics, and whether reciprocity operates differently in that tail is a question this design cannot reach. The restriction does run in a conservative direction, since the retained controls are the more answerable ones and so set a higher comparison baseline. It is also not concentrated in the newcomer stratum that carries our central result, which is dropped at close to the arm-wide

average rate. Appendix C reports the counts.

Table 3 reports the covariate balance before and after matching. Before matching, several covariates showed a meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days (SMD = 0.197) and 7 days (SMD = 0.174), reflecting that users who ask questions that receive answers tend to be more recently active. They also posted longer questions and had higher reputation (body length SMD = -0.183; owner reputation SMD = 0.115). The largest imbalance was in the tag-level accepted-answer rate (SMD = 0.695), confirming that question topic characteristics strongly predict answer receipt and underscoring the importance of matching on this variable. The propensity model addresses observable selection directly by including posting hour and day, question and title length, and owner reputation (Table 2). After matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance below 0.05 (see Table 3). This confirms adequate balance across all matching covariates; the full set is illustrated in the Love plot in Appendix B.

#### 4.5. Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates. The counting-process formulation of Andersen and Gill (1982) extends the original Cox model (Cox, 1972) to settings with time-varying covariates and recurrent events.

For each matched pair of observations (one treated, one control), we partition the observation window into intervals at the key event times: the start of the window, the posting of the question, and the arrival of the answer (actual for treated users, synthetic for controls).

We specify the hazard function as:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left( \beta_0 \cdot \text{treatment} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where  $h_0(t)$  is the baseline hazard. *treatment* is a time-constant indicator for the treatment group (1 if the question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remaining covariates are time-varying indicators that switch on at the appropriate

Table 2: Definition of Matching Covariates

| Variable name                                                                     | Definition                                                                                                                                 |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Exact matching</i>                                                             |                                                                                                                                            |
| calendarYear                                                                      | Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates. |
| topLevelTag                                                                       | First tag of the question, indicating the main topic and the community that sees the question.                                             |
| numTags                                                                           | Number of tags on the question. Treated and control questions are matched on the exact tag count, so they pose comparably scoped problems. |
| <i>Propensity score matching</i>                                                  |                                                                                                                                            |
| userTenure                                                                        | Number of days since the user’s first recorded activity on the platform, capturing cumulative experience.                                  |
| numQuestionsAskedAT                                                               | Total number of questions asked by the user before the observation window.                                                                 |
| numHelpProvidedAT                                                                 | Total number of answers provided by the user before the observation window.                                                                |
| numQuestionsAsked30D                                                              | Number of questions asked in the 30 days preceding the observation window.                                                                 |
| numHelpProvided30D                                                                | Number of answers provided in the 30 days preceding the observation window.                                                                |
| numQuestionsAsked7D                                                               | Number of questions asked in the 7 days preceding the observation window.                                                                  |
| numHelpProvided7D                                                                 | Number of answers provided in the 7 days preceding the observation window.                                                                 |
| tagAcceptShareAvg                                                                 | Average share of questions that receive an <i>accepted</i> answer within the question’s tags, capturing topic-specific difficulty.         |
| <i>Observable question and author characteristics (propensity score matching)</i> |                                                                                                                                            |
| postHour                                                                          | Hour of day at which the question was posted, capturing diurnal variation in who is online to answer.                                      |
| postDayOfWeek                                                                     | Day of week on which the question was posted.                                                                                              |
| bodyLenChars                                                                      | Length of the question body in characters, a proxy for question detail and effort.                                                         |
| titleLenChars                                                                     | Length of the question title in characters.                                                                                                |
| ownerReputation                                                                   | Asker’s reputation score at posting, capturing standing and experience beyond raw tenure.                                                  |

Table 3: Covariate Balance Before and After Matching

| Covariate            | Unmatched |         |        | Matched |         |        |
|----------------------|-----------|---------|--------|---------|---------|--------|
|                      | Tr Mean   | Ct Mean | SMD    | Tr Mean | Ct Mean | SMD    |
| userTenure           | 570.95    | 720.62  | −0.167 | 568.37  | 544.76  | 0.026  |
| numQuestionsAskedAT  | 31.11     | 26.00   | 0.059  | 30.34   | 28.04   | 0.027  |
| numHelpProvidedAT    | 15.27     | 14.15   | 0.009  | 14.36   | 12.15   | 0.018  |
| numQuestionsAsked30D | 2.00      | 1.19    | 0.197  | 1.96    | 1.84    | 0.030  |
| numHelpProvided30D   | 0.81      | 0.46    | 0.068  | 0.76    | 0.62    | 0.028  |
| numQuestionsAsked7D  | 0.57      | 0.34    | 0.174  | 0.56    | 0.53    | 0.023  |
| numHelpProvided7D    | 0.22      | 0.12    | 0.063  | 0.20    | 0.16    | 0.026  |
| tagAcceptShareAvg    | 0.51      | 0.44    | 0.695  | 0.51    | 0.51    | −0.003 |
| numTags              | 2.83      | 3.04    | −0.173 | 2.83    | 2.83    | 0.000  |
| bodyLenChars         | 1289.10   | 1658.54 | −0.183 | 1286.37 | 1338.86 | −0.026 |
| titleLenChars        | 53.80     | 57.50   | −0.172 | 53.79   | 53.97   | −0.009 |
| ownerReputation      | 4300.57   | 2541.81 | 0.115  | 4130.42 | 3499.00 | 0.041  |

*Note.* SMD = standardized mean difference. The conventional threshold for acceptable balance is  $|\text{SMD}| < 0.1$ . All post-matching SMDs fall well below this threshold.

event times. Because *treatedPostQuestion* switches on when the question is posted and remains on through the post-answer phase, the treated–control gap in the helping hazard equals  $\beta_2$  during the waiting period and  $\beta_2 + \beta_4$  after the answer arrives. The coefficient  $\beta_4$  is therefore the DiD estimate of the change at answer arrival: the shift in the treated–control gap at the time of the answer, net of any possible previous divergence. We report  $\beta_4$  as the treatment effect. We report the waiting-period coefficient  $\beta_2$  separately, as a parallel-trends indicator: because no answer has yet arrived during the waiting period, it cannot reflect a response to answer receipt. It may diverge because of anticipation or sample selection.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$\begin{aligned}
 h(t \mid \mathbf{X}(t)) = h_0(t) \exp \Big( & \boldsymbol{\beta}' \mathbf{X}(t) \\
 & + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \Big)
 \end{aligned}
 \tag{2}$$

where  $\widetilde{\log}(\text{RT})$  denotes the standardized log response-time variable (see be-

low),  $\gamma$  captures the moderation of the answer-arrival increment by answer speed, and  $\delta$  the analogous moderation during the waiting period. The logic mirrors the base model exactly. Just as  $\beta_4$  alone is the treatment effect,  $\gamma$  alone is the response-time moderation of that effect: it captures the arrival increment  $\beta_4$ , the DiD contrast on which the reciprocity effect rests. We therefore test H3 on  $\gamma$ . A negative  $\gamma$  indicates that faster answers strengthen the arrival increment, the direction H3 predicts. The waiting-period moderation  $\delta$  is the counterpart of  $\beta_2$ : it captures a pre-answer divergence that cannot reflect a response to an answer not yet received. We report it separately.

*Stratification by Tenure..* To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a non-parametric view of how reciprocity changes over the user trajectory.

*Identification..* Our identification strategy rests on the matched DiD logic embedded in the model. Propensity score matching (Rosenbaum and Rubin, 1983) ensures that treated and control observations come from users with comparable characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect.

We do not include user fixed effects in the survival models. Because we match across different users (not within a user), and because the Cox model already leverages within-observation temporal variation through its time-varying covariates (Andersen and Gill, 1982), user fixed effects are not required for identification. The DiD structure compares within-observation changes before and after the answer across treatment groups, isolating the treatment effect from stable individual differences.

The waiting-period covariate (*treatedPostQuestion*) captures any lift in helping that treated users show after asking but before receiving an answer. When this term is near zero, treated and control users move in parallel up to the answer, and the arrival increment  $\beta_4$  captures the response to receiving an answer rather than to asking a question. When it is not, the two groups diverge before treatment arrives;  $\beta_4$  still nets out the average waiting-period gap. Even a near-zero waiting-period term, however, rules out only selection effects that materialize *before* the answer. We report the waiting-period term explicitly and return to it in the Results.

## 5. Results

### 5.1. Descriptive Statistics

We analyze 24,349,760 question-level observations from 3,732,706 users on Stack Overflow, half of whose questions received at least one answer (a 50% rate fixed by construction of the matched sample, not a platform base rate). Most users help no one in a given window: they post 0.18 answers to other people’s questions on average ( $SD = 1.24$ ; median 0), so helping is rare for any individual user in any single window. Half of the users who got an answer waited less than 0.34 hours for it, though on average users wait a few hours (mean 5.08 hours,  $SD = 17.06$ ). Our users span every level of platform experience. Most questions are raised by users who had been on the platform one to three years ( $n = 6,424,968$ ), followed by those at one to six months ( $n = 4,045,952$ ) and those in their first week ( $n = 4,029,008$ ).

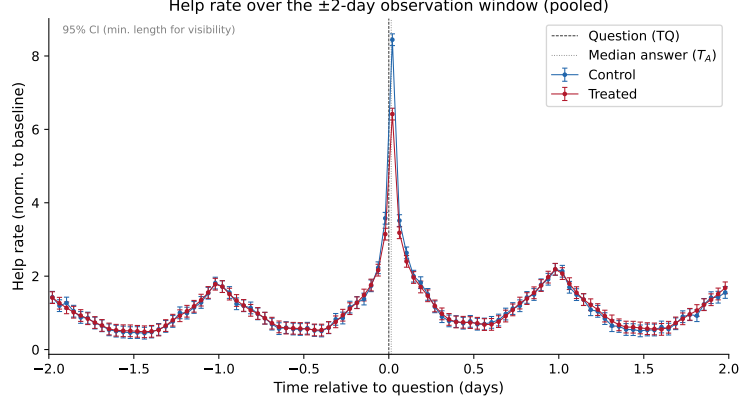
Table 4: Descriptive Statistics

| Variable                                               | Mean   | SD    | Median | N          |
|--------------------------------------------------------|--------|-------|--------|------------|
| Questions (observations)                               |        |       |        | 24,349,760 |
| Unique users                                           |        |       |        | 3,732,706  |
| % questions with answer                                | 50.00% |       |        |            |
| Help events per window (answers to others)             | 0.18   | 1.24  | 0.00   |            |
| User tenure (days)                                     | 656    | 839   | 328    |            |
| Response time (hours, treated)                         | 5.08   | 17.06 | 0.34   |            |
| <i>Response time (hours, treated) by Tenure Bucket</i> |        |       |        |            |
| < 1 Week                                               | 4.51   | 15.73 | 0.32   | 2,014,406  |
| 1 Week - 1 Month                                       | 3.59   | 13.83 | 0.26   | 775,751    |
| 1 - 6 Months                                           | 3.83   | 14.45 | 0.26   | 2,022,884  |
| 6 - 12 Months                                          | 4.24   | 15.42 | 0.28   | 1,491,553  |
| 1 - 3 Years                                            | 5.11   | 17.17 | 0.34   | 3,212,434  |
| 3 - 6 Years                                            | 6.69   | 19.87 | 0.50   | 1,807,077  |
| > 6 Years                                              | 8.66   | 22.72 | 0.75   | 850,441    |
| <i>N (unique questions) by Tenure Bucket</i>           |        |       |        |            |
| < 1 Week                                               |        |       |        | 4,029,008  |
| 1 Week - 1 Month                                       |        |       |        | 1,551,586  |
| 1 - 6 Months                                           |        |       |        | 4,045,952  |
| 6 - 12 Months                                          |        |       |        | 2,983,180  |
| 1 - 3 Years                                            |        |       |        | 6,424,968  |
| 3 - 6 Years                                            |        |       |        | 3,614,184  |
| > 6 Years                                              |        |       |        | 1,700,882  |

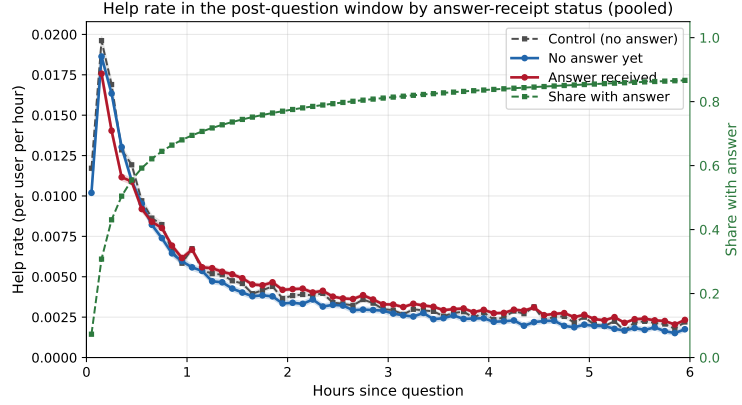
**Notes:** Help events per window are raw event counts. Estimation applies a one-hour interval rounding (see Implementation Details).

### 5.2. Descriptive Evidence: Help Rates and the Parallel Trends Assumption

Panel (a) of Figure 3 plots the normalized help rate for treated and control users over the two days before and after question posting. Before the question ( $t < T_Q$ ), treated and control help rates track each other closely, consistent with the parallel-trends assumption and with the effectiveness of propensity score matching. At the time of posting, help rates rise sharply in both groups because users are actively engaged on the platform. The spike is more pronounced for controls, who continue waiting for an answer, than



(a) Treated versus matched control users over a fixed  $\pm 2$ -day window centered on the question. The dashed line marks the question time ( $T_Q$ ), the dotted line the median answer arrival; error bars are 95% confidence intervals.



(b) Treated users in the post-question window, split by whether their answer has yet arrived, against matched controls. The right axis gives the cumulative share of treated users answered by each hour.

Figure 3: Raw help rates over the observation window, pooled across all tenure buckets. Panel (a) bears on the parallel-trends logic of the design (Section 5.2) and uses a symmetric question-centered axis ( $T_Q \pm 2D$ ); panel (b) bears on the re-engagement reading of the response-time pattern (Section 5.5). Appendix D reproduces both panels by tenure bucket.

for treated users, whose questions are typically answered within the hour. Beyond this mechanical spike, the treated–control differences implied by our estimates are too small to be visible in raw normalized rates. We return to the control group’s larger spike at the time of the question when discussing the response-time pattern.

### 5.3. Generalized Reciprocity Effect (H1)

To test Hypothesis 1, we estimate the pooled Cox model across all experience levels (Table 5). The DiD treatment effect is the increment in the helping hazard at answer arrival ( $\beta_4$ ). We report the waiting-period coefficient ( $\beta_2$ ) alongside it as a parallel-trends diagnostic. Receiving an answer is associated with a modest increase in subsequent helping. The arrival increment corresponds to  $HR = 1.049$  (95% CI [1.044, 1.054],  $p < .001$ ), equivalent to a 4.9% higher helping hazard after answer receipt. The waiting-period coefficient is small and negative ( $\beta_2 = -0.007$ ,  $p = .01$ ), substantially smaller than the arrival increment. Treated users therefore show no upward divergence from matched controls before answer arrival. This pattern reduces concern that the estimated treatment effect reflects a pre-existing trend in helping behavior. Appendix Table F.12 reports matched-pair bootstrap confidence intervals that preserve the matched structure and support the same conclusion. Overall, the results support Hypothesis 1, although the estimated effect is smaller than those reported in prior observational studies.

Table 5: Pooled Cox Regressions (All Experience Levels)

|                                                                                   | Main                  | Main + Speed           |
|-----------------------------------------------------------------------------------|-----------------------|------------------------|
| <i>Treatment effect (DiD): increment at answer arrival (<math>\beta_4</math>)</i> |                       |                        |
| Answer arrival ( $\times$ Post-Answer Received)                                   | 0.0479***<br>(0.0026) | 0.0373***<br>(0.0022)  |
| Hazard Ratio [95% CI]                                                             | [1.044, 1.054]        | [1.034, 1.042]         |
| Hazard Ratio [bootstrap 95% CI]                                                   | [1.032, 1.075]        | —                      |
| <i>Parallel-trends diagnostic: waiting period (<math>\beta_2</math>)</i>          |                       |                        |
| Waiting period ( $\beta_2$ ): Received Answer $\times$ Post-Question              | -0.0065*<br>(0.0026)  | 0.0142***<br>(0.0021)  |
| <i>Response Time Interaction</i>                                                  |                       |                        |
| Answer arrival $\times$ log(RT) ( $\gamma$ )                                      | —                     | -0.0636***<br>(0.0015) |
| Waiting period $\times$ log(RT) ( $\delta$ )                                      | —                     | 0.0630***<br>(0.0013)  |
| N (matched question rows)                                                         | 24,349,760            | 24,349,760             |
| Person-interval rows                                                              | 62,379,420            | 62,379,420             |
| Events                                                                            | 3,435,973             | 3,435,973              |

**Notes:** Matched-pair bootstrap 95% CI for the pooled arrival increment is from Table F.12.

N counts matched question-level rows (treated + control). The Cox partial likelihood is fit over the person-interval rows expanded from them. The point estimates use the full interval set; only the matched-pair bootstrap replicates (Table F.12) draw an 8M-interval-row subsample per replicate.

Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ;  $^\dagger p < .1$

In response to the concern that users do not only reciprocate with answers but through other lower-cost contributions to the platform, Table 6 re-estimates the pooled model under broader outcome definitions. The decomposition confirms that the reciprocity signal is concentrated in *answers*. Comments alone show only a slight elevation (arrival HR = 1.01, 95% CI [1.01, 1.02]), well below the answers-only increment, and adding comments to answers dilutes rather than amplifies the estimate (answers-plus-comments arrival HR = 1.02, [1.02, 1.03]). Edits of other users' posts carry a larger association (edits-only arrival HR = 1.16), but we report them separately because they reflect curation more than costly contribution. When we combine all channels, the figure sits just around the answers-only estimate (arrival HR = 1.05). The waiting-period diagnostic is near zero in every column ex-

cept edits ( $\beta_2 = +0.06$ ). The answers-only estimate we use is thus a meaningful high-cost channel to reciprocate, while it looks like many are also increasing their low-cost contributions through edits as well.<sup>1</sup>

Table 6: Reciprocity Effect Decomposed by Help Type

| Outcome                    | Arrival HR [95% CI] ( $\beta_4$ ) | Waiting $\beta_2$ | N          | Events    |
|----------------------------|-----------------------------------|-------------------|------------|-----------|
| Answers only               | 1.049[1.044, 1.054]               | -0.01             | 24,349,760 | 3,435,973 |
| Comments only              | 1.013[1.007, 1.018]               | +0.01             | 24,349,760 | 2,170,711 |
| Edits only                 | 1.163[1.155, 1.170]               | +0.06             | 24,349,760 | 992,369   |
| Answers + comments         | 1.024[1.020, 1.029]               | -0.00             | 24,349,760 | 5,025,482 |
| Answers + comments + edits | 1.045[1.041, 1.050]               | -0.00             | 24,349,760 | 5,679,602 |

**Notes:** Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

#### 5.4. Experience Moderation of Reciprocity (H2)

To test Hypothesis 2, we estimate the model separately within each of the seven tenure buckets (Table 7; Figure 4). The response is strongest at entry and fades to nothing among the most experienced. The decline across the middle of the tenure range is not strictly monotone. The newest users respond most: in their first week, users who received an answer went on to help roughly 7.1% more often than their matched counterparts (arrival HR = 1.071, 95% CI [1.056, 1.086],  $p < .001$ ). The response then dips in the one-week-to-one-month bucket (HR = 1.03 [1.01, 1.05]), recovers to a plateau of about 1.05 that holds from one month through three years, and weakens again at three to six years (HR = 1.02 [1.01, 1.04],  $p < .001$ ). Only users with more than six years on the platform show no response at all (HR = 1.00 [0.98, 1.02], n.s.). Read at this level, the pattern supports Hypothesis 2: reciprocity is a newcomer phenomenon that fades with experience. We do not read the one-week-to-one-month dip as substantively meaningful. Because the

<sup>1</sup>Editing others' posts skews strongly toward experienced users, because editing them directly requires 2,000 reputation; below that threshold an edit must be submitted as a suggestion and cleared in peer review before it is recorded. Users who edit during their window are accordingly far more senior than those who answer (median reputation 8,032 against 2,918, and 329 across all windows), and only 2.6% of the windows containing an edit belong to users in their first week, against 16.8% of windows overall. The channel is not formally closed to newcomers — 24.3% of editing windows fall below the 2,000-reputation threshold, reflecting the suggested-edit route — but this estimate is not driven by them.

newcomer stratum is one of our central findings, we subject the treatment effect to a matched-pair bootstrap that resamples whole pairs and preserves the matched structure. At 500 replicates, the bootstrap 95% CI is [1.06, 1.08], confirming the model-based interval (Appendix Table F.12).

For newcomers, both the arrival increment ( $\beta_4$ ) and the waiting-period term ( $\beta_2$ ) are positive and statistically significant ( $\beta_4 = 0.068$ ,  $\beta_2 = 0.050$ , both  $p < .001$ ), whereas among veterans both are near zero. The positive waiting-period term means that newcomers begin helping more even before their answer arrives. The arrival increment doubles the pre-trend divergence, but a nonzero pre-trend is a caution: the design cannot fully separate that pre-answer divergence from reciprocity (Section 6.5).

Stack Overflow was substantially disrupted by the release of generative-AI tools in late 2022, which sharply reduced human question-and-answer activity (del Rio-Chanona et al., 2024; Burtch et al., 2024). Our own data carry a trace of it: the per-window helping rate is markedly lower after 2020 than before, about 0.057 against 0.164 helping events per matched window. That disruption is not the focus of this paper, but it invites a natural check: do our findings hold before and after it? Because calendar year enters the design as an exact-matching stratum, we can re-estimate our two central results, the pooled effect (H1) and the tenure gradient (H2), separately by era (Appendix Table G.13).

The pooled effect is stable across the divide: the arrival increment is 1.04 on the pre-2020 subsample and 1.04 on the disjoint post-2020 data (95% CI [1.03, 1.06]), against 1.05 in the full sample. The newcomer effect is not, however. It is present before the generative-AI era (pre-2020 arrival HR = 1.07, matching the full sample) but not distinguishable from one afterward (post-2020 HR = 1.00, [0.97, 1.03]), on far fewer events (31,814 versus 336,966). We read this as a genuine difference between periods: the newcomer response is well identified in the pre-generative-AI data but appears to weaken after 2020. We develop possible explanations for why it weakens in the discussion (Section 6.6).

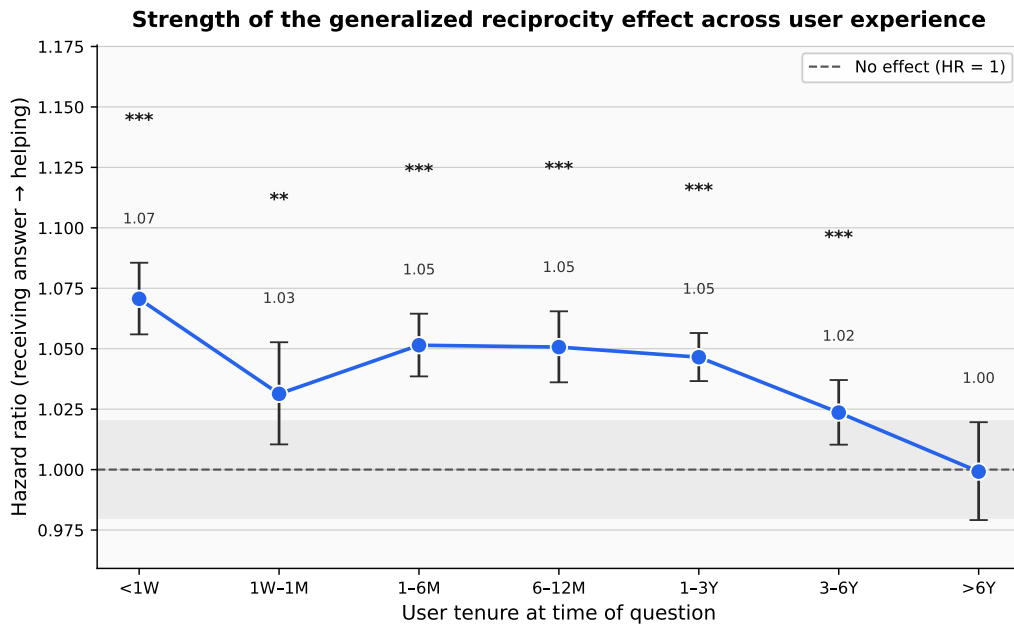


Figure 4: Strength of the generalized reciprocity effect across user experience. Each point shows the arrival-increment hazard ratio ( $\beta_4$ ) estimated separately within a tenure bucket; error bars indicate 95% confidence intervals. Stars denote significance (\*\* $p < .01$ ; \*\*\* $p < .001$ ).

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

|                                                                             | < 1<br>Week    | 1 Wk–<br>1 Mo  | 1–6<br>Mo      | 6–12<br>Mo     | 1–3<br>Yr      | 3–6<br>Yr      | > 6<br>Yrs     |
|-----------------------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| <i>Treatment effect: increment at answer arrival (<math>\beta_4</math>)</i> |                |                |                |                |                |                |                |
| Answer<br>arrival ( $\times$<br>Post-Answer<br>Received)                    | 0.0683***      | 0.0309**       | 0.0502***      | 0.0494***      | 0.0455***      | 0.0233***      | -0.0008        |
|                                                                             | (0.0071)       | (0.0105)       | (0.0063)       | (0.0071)       | (0.0048)       | (0.0067)       | (0.0103)       |
| <i>Hazard Ratio</i><br>[95% CI]                                             | [1.056, 1.086] | [1.010, 1.053] | [1.039, 1.064] | [1.036, 1.065] | [1.037, 1.056] | [1.010, 1.037] | [0.979, 1.020] |
| <i>Waiting period (<math>\beta_2</math>, parallel-trends diagnostic)</i>    |                |                |                |                |                |                |                |
| Received                                                                    | 0.0504***      | -0.0266*       | -0.0147*       | -0.0062        | 0.0000         | -0.0107        | -0.0167†       |
| Answer $\times$<br>Post-Question                                            | (0.0070)       | (0.0105)       | (0.0063)       | (0.0072)       | (0.0049)       | (0.0066)       | (0.0101)       |
| N                                                                           | 4,029,008      | 1,551,586      | 4,045,952      | 2,983,180      | 6,424,968      | 3,614,184      | 1,700,882      |
| Events                                                                      | 368,780        | 238,189        | 694,001        | 537,263        | 1,037,470      | 428,036        | 132,234        |

**Notes:** N = unique questions (treated + control) in the Cox sample.

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled arrival increment ( $\beta_4$ ) are in Table F.12.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; † $p < .1$

### 5.5. Response Time Moderation (H3)

To test Hypothesis 3, which predicted that faster responses strengthen the reciprocity effect, we estimate the arrival increment and its response-time moderation separately by tenure bucket (Table 8); Table 9 and Figure 5 give a more flexible discrete-bin view, and the figure sets each baseline bin beside its answer-quality-controlled counterpart. The evidence supports the speed prediction. The continuous log-response-time interaction with the arrival increment ( $\gamma$ ) is negative in every tenure bucket and statistically significant in all but the most senior: users who waited less for their answer responded more strongly, the direction H3 predicts. The moderation is strongest early in the trajectory (about  $-0.07$  to  $-0.09$  from the newcomer through one-year buckets), weakens with tenure ( $-0.05$  at one to three years,  $-0.02$  at three to six), and is indistinguishable from zero among users with more than six years.

The discrete bins tell a consistent, if noisier, story. Users who receive help within 30 minutes barely respond at all, while those who receive help between 30 and 60 minutes respond the strongest (1.19 [1.17, 1.21]). It declines toward one over the following day (1.06 at one to two hours, 1.02–1.05 through one day), with a smaller secondary elevation at one to three days (1.09 [1.07, 1.11]). The pronounced early peak shows a sweet-spot pattern; we return to explanations for it below.

The waiting-period gap carries a response-time moderation of its own ( $\delta$ ), positive in every bucket — the counterpart of the  $\beta_2$  diagnostic, capturing how the *pre-answer* treated–control gap varies with answer speed. Because no answer has arrived yet, it reflects anticipation or selection. It is separate from  $\gamma$ , the moderation of the arrival increment: the DiD nets the pre-answer gap out of the increment, so the moderation we report ( $\gamma$ ) reflects the change at answer arrival, not treated users’ pre-answer trajectories.

A natural concern is that response time reflects unobserved answer quality. People who answer within minutes tend to post something terse or link-only, whereas those who take longer tend to write fuller, more detailed answers. So, what prompts users to reciprocate may be the quality of the answer they received, not its speed. If quality and response time are correlated in this way, the sweet-spot pattern could reflect quality differences rather than speed. We probe the threat with two checks.

Table 8: Response Time Moderation of the Reciprocity Effect

|                                                                                                                  | < 1<br>Week         | 1 Wk–<br>1 Mo | 1–6<br>Mo  | 6–12<br>Mo | 1–3<br>Yr  | 3–6<br>Yr  | > 6<br>Yrs |
|------------------------------------------------------------------------------------------------------------------|---------------------|---------------|------------|------------|------------|------------|------------|
| <i>Treatment effect: arrival increment (<math>\beta_4</math>, at mean response time)</i>                         |                     |               |            |            |            |            |            |
| Answer<br>arrival (net<br>post-answer)                                                                           | 0.0101 <sup>†</sup> | 0.0178*       | 0.0348***  | 0.0417***  | 0.0501***  | 0.0353***  | 0.0139     |
|                                                                                                                  | (0.0057)            | (0.0086)      | (0.0052)   | (0.0060)   | (0.0041)   | (0.0056)   | (0.0086)   |
| <i>Response-time moderation (<math>\gamma</math> = arrival; <math>\delta</math> = waiting-period diagnostic)</i> |                     |               |            |            |            |            |            |
| Answer<br>arrival $\times$<br>log(RT) ( $\gamma$ )                                                               | -0.0704***          | -0.0778***    | -0.0910*** | -0.0863*** | -0.0547*** | -0.0228*** | -0.0015    |
|                                                                                                                  | (0.0048)            | (0.0068)      | (0.0039)   | (0.0043)   | (0.0028)   | (0.0036)   | (0.0055)   |
| Waiting<br>period $\times$<br>log(RT) ( $\delta$ )                                                               | 0.0236***           | 0.0725***     | 0.0892***  | 0.0829***  | 0.0670***  | 0.0453***  | 0.0278***  |
|                                                                                                                  | (0.0039)            | (0.0060)      | (0.0035)   | (0.0038)   | (0.0024)   | (0.0031)   | (0.0045)   |
| N                                                                                                                | 4,029,008           | 1,551,586     | 4,045,952  | 2,983,180  | 6,424,968  | 3,614,184  | 1,700,882  |
| Events                                                                                                           | 368,780             | 238,189       | 694,001    | 537,263    | 1,037,470  | 428,036    | 132,234    |

**Notes:**  $\gamma$  and  $\delta$  enter response time linearly in standardized log hours; the flexible discrete-bin counterpart of this specification is Table 9 and Figure 5.

Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; <sup>†</sup> $p < .1$

Table 9: Treatment Effect by Response-Time Bin

| Response time | HR       | 95% CI         | N          | Events    |
|---------------|----------|----------------|------------|-----------|
| 0-15 min      | 0.998    | [0.989, 1.007] | 17,412,740 | 2,484,407 |
| 15-30 min     | 1.004    | [0.989, 1.019] | 13,934,456 | 1,812,023 |
| 30-60 min     | 1.190*** | [1.174, 1.206] | 13,549,736 | 1,744,784 |
| 1-2 hr        | 1.056*** | [1.040, 1.072] | 13,211,144 | 1,698,505 |
| 2-4 hr        | 1.030*** | [1.013, 1.047] | 12,958,927 | 1,663,748 |
| 4-8 hr        | 1.017†   | [0.999, 1.035] | 12,762,008 | 1,639,963 |
| 8-12 hr       | 1.019    | [0.995, 1.044] | 12,464,162 | 1,601,705 |
| 12-24 hr      | 1.048*** | [1.028, 1.067] | 12,645,932 | 1,626,049 |
| 1-3 days      | 1.088*** | [1.069, 1.108] | 12,592,667 | 1,635,906 |
| >3 days       | 1.015    | [0.992, 1.038] | 12,391,574 | 1,628,425 |

**Notes:** Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool.

Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; † $p < .1$

The first is the direct test for the response-time pattern: we re-estimate each response-time bin with the three answer-quality controls, which Figure 5 sets beside the baseline fits in panel (b) and Appendix Table G.17 reports in full. This adjustment is deliberately aggressive. Vote score and acceptance are post-treatment, and acceptance in particular requires the asker to return to the platform, so conditioning on them absorbs some of the effect itself, not only the quality confound. The clearest sign of this over-adjustment is that under the controls the fastest bins fall significantly *below* one (0.95), which has no plausible other explanation. Even so, the 30–60-minute peak falls only from 1.19 to 1.03 (95% CI [1.02, 1.04],  $p < .001$ ) and remains the significant early-window maximum, and the one-to-three-day elevation persists (1.09). The sweet-spot pattern remains even after an adjustment that likely underestimates the true causal effect.

The second check concerns the pooled *level* of the arrival increment. Appendix Table G.15 reports how it shifts as the same controls enter one at a time. Answer length, which is the least endogenous signal and plausibly correlated with answer quality, barely moves it (from 1.04 to 1.03). The more endogenous vote-score and acceptance signals pull it further, to 1.01 and then below one (0.98). Re-coding the vote score to account for its heavy right tail leaves the below-one full-specification endpoint intact (0.98 under all three codings), but not the intermediate 1.01 step, which is indistinguishable from one under the signed-log and binned codings; we therefore do not read that

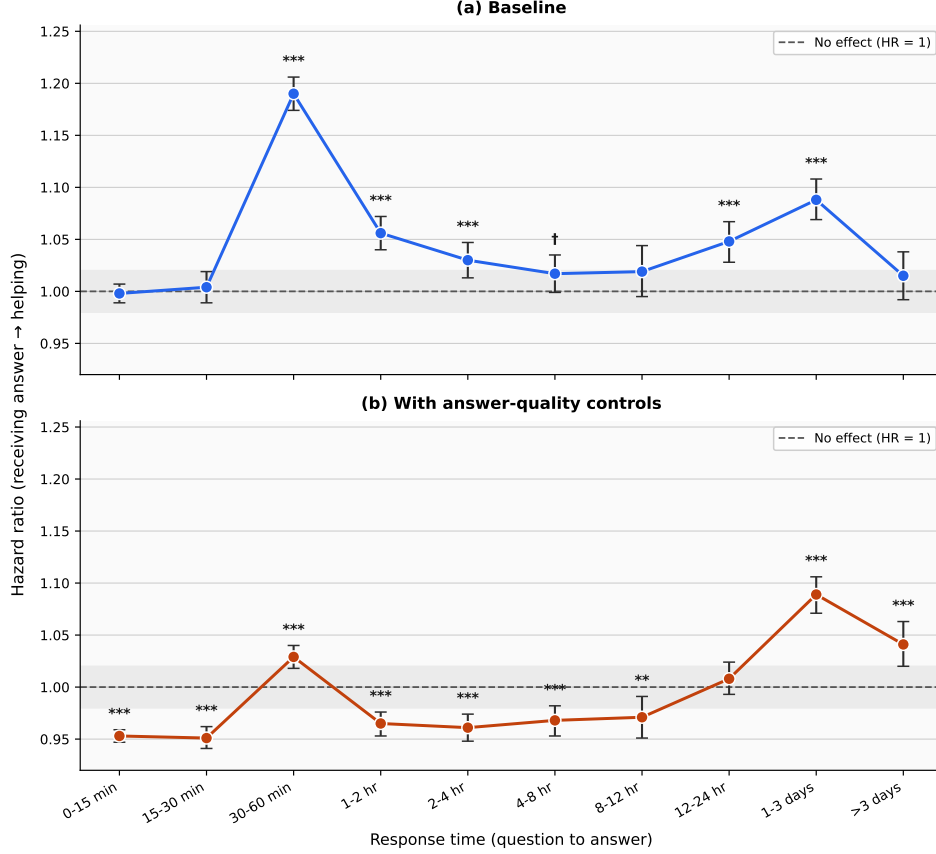


Figure 5: Treatment effect by response-time bin, pooled across all tenure buckets. Each point is the arrival-increment hazard ratio ( $\beta_4$ ) estimated within that response-time bin, relative to the matched control; error bars indicate 95% confidence intervals and the dashed line marks no effect ( $HR=1$ ). Panel (a) gives the baseline fits of Table 9. Panel (b) refits every bin with three answer-quality controls, namely acceptance, first-answer score, and answer length, reported in full in Appendix Table G.17. Because the panels share both axes, the adjustment can be read off bin by bin: the 30–60-minute peak shrinks from 1.19 to 1.03 while remaining the early-window maximum, the one-to-three-day elevation is untouched, and the two fastest bins fall below one. Section 5.5 reads those sub-one estimates as a sign that the adjustment is over-aggressive rather than that the effect is absent, since acceptance and vote score are post-treatment.

step as evidence of an above-one effect (Appendix Table G.18). As above, these level estimates are bounds, not estimates of the effect net of quality: conditioning on post-treatment signals likely takes away from the effect itself. Our quality measures are coarse, so we cannot cleanly separate the effects of timing and quality, and cannot fully rule out that answer quality, imperfectly measured, explains the response-time pattern.

One interpretation of the sweet-spot pattern is *re-engagement*: users who post a question may remain actively engaged while waiting during the ongoing session on the platform, elevating helping during the waiting period regardless of whether an answer arrives. On this account, a fast answer might lead users to end their session and, with it, their engagement without returning to the platform. If an answer arrives after they have ended their session, they might return to review it and reciprocate in the newly started session triggered by the answer. Later answers are more likely to re-engage the user in another session, which answers in the same initial session do not.

Panel (b) of Figure 3 provides descriptive context, comparing users who have received an answer (red) with users who will receive one but are still waiting (blue), alongside matched controls. In the first half-hour after the question — when only the fastest answers have arrived — users still waiting help at the highest rate, consistent with the idea that users help while waiting on the platform. From roughly an hour onward, the ordering flips: users who have received an answer help at a modestly higher rate than those still waiting, consistent with a small post-answer elevation. We read this as descriptive evidence for the re-engagement mechanism. Appendix D shows the same qualitative pattern in every tenure bucket. These raw-rate orderings are descriptive: the differences are small, and answer-arrival timing is not random within the treated group, so we do not read them as decisive evidence for any one mechanism. Still, they are consistent with the modest reciprocity effect and the re-engagement mechanism the models estimate. Overall, however, re-engagement is speculative since we do not observe session boundaries (logins, logouts, inactivity gaps). We return to this question in Section 6.3.

## 6. Discussion

### 6.1. Methodological Contribution

Researchers who study generalized reciprocity from platform data face a persistent difficulty: generally active users are more likely to ask questions

and post answers. A correlation between help received and help given might therefore reflect nothing more than how active a user is. Prior observational studies that report positive associations without accounting for this selection (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006) may have overstated the role of reciprocity.

Our matched DiD survival design addresses that confound. It compares changes in helping, before and after a question is posted, between users whose questions received answers and propensity-score-matched users whose questions did not. The staggered adoption design and modeling the waiting period separately allow us to identify the generalized reciprocity effect associated with answer arrival. The estimate this yields is small and strongest among newcomers, and it is substantially smaller than many earlier reports, which suggests that these previous estimates were confounded by general differences in user activity.

The design travels past this particular question. Any platform event that arrives at a time the researcher does not control, such as a first review, a moderator intervention, a badge, or a reply, raises the same confound, and the same comparison of matched users before and after the event addresses it. Our design is therefore a general contribution that researchers and platform designers can use to study behavioral interdependencies between different user-driven platform events.

### *6.2. Reciprocity as a Contributor-Recruitment Mechanism*

Users respond to being helped less the longer they have been on the platform. The newest respond most ( $HR = 1.07$ ) and the most experienced not at all (arrival  $HR = 1.00$ ; Table 7). Read against the theory, this gradient does more than qualify the reciprocity account. It relocates it. Generalized reciprocity on the platform we study is not the general driver of cooperation among members that the literature often takes it to be. It is mostly a channel that works at the point of entry and weakens as users settle in.

There are several plausible theoretical accounts for this effect. The norm-displacement account developed in Section 2.3 offers one answer: as users gain experience, the community’s own incentives crowd out a general reciprocal impulse. Experienced users on Stack Overflow have reasons to help that newcomers do not yet have — reputation, visible status, a stake in the community. That the most experienced users do not respond at all fits this account. Our design tracks what users did rather than what moved them, so we cannot confirm it.

The reframing also makes sense of an inconsistent literature. Researchers who sample mostly experienced users, or who pool across experience levels dominated by experienced contributors, should find weaker reciprocity effects. Those who catch users early should find stronger ones. Prior findings may diverge, then, not because anyone erred, but because reciprocity is not a fixed property of a population.

What our data does not tell us is what takes reciprocity’s place. Our analysis establishes that generalized reciprocity declines with experience; it does not characterize the motives that replace it. The theoretical argument implicates community-specific norms and status concerns, but testing that account needs direct measures of them — surveys of experienced contributors, or experiments that manipulate reputation visibility and observe its effect on helping.

### *6.3. The Response-Time Pattern and Its Interpretation*

The response-time analysis tested a prediction drawn from laboratory studies of gratitude: that faster help produces stronger reciprocity. That much fits a speed account of gratitude, in which help arriving while the need is still felt draws a stronger response. The evidence supports it across the user trajectory. We also find, however, that users who receive an answer immediately (within 30 minutes) show no reciprocity. Instead, users who wait for their answers a little show the strongest reciprocity. A pure speed account gives no reason why extremely fast answers should do nothing at all.

Three readings of that anomaly compete, and we cannot cleanly tell them apart with our data. The first is answer quality: the quickest replies may also be the thinnest, which the Results adjudicate as far as the dump allows (Section 5.5). The second is re-engagement: an answer that lands while a user is still on the site may end an ongoing help session, while one that lands later brings them back and may trigger a new help session. The third is question characteristics: response time stands in part for properties of the question that the tag-level accepted-answer rate does not capture, since questions that draw fast answers may differ from those that wait. All three are conjectures. We observe neither session boundaries nor question difficulty, and our quality signals are coarse.

The pattern can also speak to the norm-displacement account. Among the most experienced users, the arrival increment and its response-time moderation are both near zero, which is consistent with the mechanism having

faded rather than merely weakened. While this needs to be treated with caution, this finding gives further credibility to our theoretical account.

Settling the interpretation needs data the archive does not hold. Session-level records would test re-engagement directly. Sophisticated, text-based measures of question complexity or specificity would separate timing from difficulty. Randomly assigned answer timing would separate both from speed itself. Each of these may be available to a platform or an experimenter, offering the opportunity to build on this work.

#### *6.4. Implications for Platform Design*

Because the response we observe is modest and strongest among newcomers, designers can use it mainly to recruit new contributors rather than to instigate continuous contributions from established members.

Designers who want to grow their contributor base should therefore make sure that new users get answers to their questions. Newcomers who received an answer went on to help about 7% more often than their matched counterparts ( $HR = 1.07$ ; Table 7), or roughly 8 additional helping events per 1,000 answered newcomer questions (Table 10). The absolute figure is small because helping is rare in this population to begin with: matched newcomer controls average about 0.11 helping events per observation window, and most control observations record none at all. Put differently, about 125 newcomer questions must be answered to produce one additional answer post. This is still the highest yield of any tenure group, next to the statistically indistinguishable one-to-twelve-month strata (roughly 126–128 questions) but clearly more favorable than the pooled figure of about 158. The case for concentrating on newcomers therefore rests both on their being the group that responds most and on a per-answer yield at least as favorable as any other group’s. The most experienced users do not respond (arrival  $HR = 1.00$ ), so a designer who wants to maximize reciprocated answers should prioritize their questions lower when directing attention across questions. Designers thus have most leverage with reciprocity at the point of entry.

Table 10: Absolute Effect Sizes and NNT

| Row                  | Control rate | HR    | ARD      | Events/1k | NNT   | NNT CI      |
|----------------------|--------------|-------|----------|-----------|-------|-------------|
| < 1 Week             | 0.1133       | 1.071 | 0.00800  | 8.00      | 125.0 | 103.2–157.8 |
| 1 Week - 1 Month     | 0.1401       | 1.031 | 0.00439  | 4.39      | 227.8 | 135.5–684.6 |
| 1 - 6 Months         | 0.1542       | 1.051 | 0.00793  | 7.93      | 126.1 | 100.6–168.2 |
| 6 - 12 Months        | 0.1542       | 1.051 | 0.00782  | 7.82      | 127.9 | 99.0–179.6  |
| 1 - 3 Years          | 0.1370       | 1.047 | 0.00637  | 6.37      | 157.0 | 129.2–199.4 |
| 3 - 6 Years          | 0.1032       | 1.024 | 0.00244  | 2.44      | 410.7 | 261.5–939.9 |
| > 6 Years            | 0.0716       | 0.999 | -0.00006 | -0.06     | —     | —           |
| Pooled (all tenures) | 0.1287       | 1.049 | 0.00631  | 6.31      | 158.4 | 142.9–177.5 |

**Notes:** Based on the answer-arrival increment ( $\beta_4$ ). *Control rate* is the mean helping events per observation window among matched control questions.

ARD = control rate  $\times$  (HR – 1) treats the hazard ratio as a rate ratio (a first-order approximation); *Events/1k* =  $1000 \times$  ARD; and NNT =  $1/\text{ARD}$  is an approximate “answers needed per additional helping event,” not a true number-needed-to-treat.

Response speed is a weaker basis for design than it first appears. Users who received an answer more quickly generally responded more strongly, and those answered between 30 and 60 minutes responded the most (Tables 8 and 9). The obstacle is not the size of the pattern but its interpretation: it admits two readings that point designers in different directions. Instead of specific interventions, we therefore propose designers two questions to run on their own platforms.

The first question is how answers land in a user’s session. Platforms record when a user arrives, how long they stay, and whether an answer reaches them inside that visit or after it has ended. If the sweet spot reflects re-engagement, an answer that arrives mid-visit competes with the activity a user is already engaged in, while one that arrives after the visit gives them a reason to come back. Designers would then be working on the return trip. What matters is when notifications are sent to the user, and how prominently they are surfaced during potential help sessions. Time to first answer would be a suboptimal metric to optimize.

The second question is whether fast answers are *low quality* and clear the queue without doing much for the asker. The two fastest bins are indistinguishable from no effect (0.998 and 1.004; Table 9), which is what we would expect if the quickest replies are also the thinnest. A newcomer whose first exchange with the community is a curt reply may simply not return. If low-quality answers drive the fastest bin, the design problem sits in quality at the

fast end rather than in delay, and platforms would do better to discourage quick throwaway replies particularly to newcomers. Our own checks speak against this reading: the 30–60-minute peak survives aggressive quality controls (Section 5.5). The quality signals in the public dump are too coarse to settle it, but platforms that have high-fidelity measures of answer quality and newcomer retention can run this analysis.

Both readings lead to the same caution. Designers stand on firmer ground making sure that newcomers’ questions get answered at all, and that users notice when they do, than tuning for a particular response window.

### 6.5. *Limitations and Future Research*

Our study has several limitations in what we measure, what we identify causally, what we observe, and where our findings apply. We discuss them one by one alongside opportunities for future research.

Start with what we measure. Our helping outcome is the posting of an answer to another user’s question, which we treat as a costly public contribution. Reciprocity can also run through channels we do not observe. On the platform, upvotes are the clearest case: the public dump does not include the voter’s identity, so votes given cannot be attributed to a user at all. Off the platform, a helped user might answer a colleague directly, write the solution up elsewhere, or pass it on somewhere Stack Overflow never records. If newcomers reciprocate through these unobserved channels more than veterans do, our estimates miss more of their reciprocity than of the veterans’, which makes the reported tenure finding a conservative one.

Turn next to what we can identify. Our estimate of reciprocity is the arrival increment, the change in helping at the moment an answer arrives. It is identified under several assumptions, most critically, that nothing else which raises helping changes at that same moment.

Three kinds of selection bear on that assumption. The matching handles selection on *observables*, the characteristics that predict getting answered, and leaves every covariate below conventional balance thresholds (Figure B.7). The DiD removes selection on *stable unobservables*. The third kind remains: something that changes about the user inside the window and affects answer arrival and subsequent helping at once. Someone who becomes more active may draw an answer sooner and, for the same reason, help more. We think this assumption is defensible because the waiting-period term catches everything that already surfaces before the answer.

However, if any behavioral change (that caused the answer) affects helping only after the moment of answer receipt, we cannot separate it from reciprocity. Fully settling it needs a design that makes answer arrival exogenous: a field experiment that seeds answers to a random subset of questions, or a platform change that shifts answering capacity for reasons unrelated to who is asking.

That brings us to what we cannot see. The design captures a behavior change but not the psychology behind it. Gratitude, moral obligation, social expectation, incentive displacement, and re-engagement are all *proposed* interpretations, each consistent with part of the pattern and none identified by our data. If social expectation is the behavioral mechanism, users who feel more visible after being answered should be helping more. Testing visibility as a mechanism needs a measure of how visible a question was at the time, which the dump does not provide. Beyond that, we observe no session boundaries, no reputation salience, and no emotional states. Which of gratitude, moral obligation, and social expectation actually moves users in a setting like this one is not a question our data can settle, and we leave it to psychologists with the instruments to measure it. Separating these pathways calls for measures taken while people are on the platform, such as experience sampling of emotional state. Interventions that target the proposed psychological mechanisms one at a time offer an alternative test.

Finally, where the estimates apply. Stack Overflow is a large, relatively anonymous, gamified platform built around technical question-answering. The patterns we estimate may not transfer to smaller or relation-based communities such as enterprise Slack channels, academic peer review, or intimate hobby forums, where social ties and non-status motives differ. We therefore refrain from claiming that generalized reciprocity operates on platforms with different dynamics and mechanisms of social engagement. Our estimates describe one platform, and largely its pre-generative-AI era.

### *6.6. Generalized Reciprocity in the Generative-AI Era*

Our analysis window runs from Stack Overflow’s inception through 1 April 2025. Because calendar year is an exact-matching stratum, the pre-2020 period supplies most of the analysis, which dates our results to a platform where community answering was still the norm. The era split reported in Section 5.4 sharpens the point: the pooled effect survives the release of ChatGPT, while the newcomer response does not. The recruitment channel

we identify, in other words, weakened at about the point the platform began to change.

Why it weakened is a question our data cannot answer, but we offer two conjectures. One is composition: the newcomers who still join after ChatGPT’s release are a selected group, and the post-2022 user drain was strongest among the junior users who carried the earlier response (Burtch et al., 2024). The other is opportunity: activity declined most in the topics where generative AI tools perform best (del Rio-Chanona et al., 2024), so the small volume of questions that remain may be harder, and newcomers compete with a relatively large group of tenured users in answering them, leaving a newly helped newcomer with fewer questions they can answer. Both would weaken the recruitment channel because the composition of users and questions on the platform changed and not because of reciprocity itself. Neither is testable here, since we observe neither question difficulty over time nor who chose not to join.

The larger question is what becomes of reciprocal recruitment when a machine supplies the answer. If users increasingly get help from an assistant rather than a person, the human exchange that gave them their modest push into contributing may disappear. The demand for human contribution does not disappear with LLM-based chatbots, however. Systems still depend on people for correction, feedback, and the ground-truth detail that populates shared resources, such as local detail added to mapping platforms, which LLMs cannot infer on their own and may actually be trained on. Whether generalized reciprocity can recruit those contributors when the help that triggers it comes from an LLM is a question our study design can be used to answer.

## 7. Conclusion

Generalized reciprocity is widely credited with sustaining cooperation on knowledge-sharing platforms, yet observational tests run into a stubborn confound: generally active users ask and answer more questions at the same time. Our matched DiD survival design confronts that confound directly and delivers a cleaner test of whether people help more once their own questions are answered. The response to being helped is strongest among newcomers and fades to nothing among the longest-tenured users. On Stack Overflow, generalized reciprocity works less as a general engine of cooperation than as a channel for recruiting new contributors. This continues to matter as

generative-AI systems absorb routine question-answering. People still supply what machines cannot generate on their own: feedback, input, correction, and ground-truth knowledge. For much of this, these systems in fact depend on the people who provide it. Understanding how such contributors are recruited therefore remains important. Our study contributes both a finding and a framework: the leverage of generalized reciprocity is greatest among newcomers, and our design offers a general approach to understanding the causal effect of platform events on user behavior, and with it to measuring and strengthening user contributions where they matter most.

## **Funding**

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## **Ethics statement**

This study analyzed de-identified, publicly available archival records from Stack Overflow. No interaction with human participants occurred, and no personally identifiable information was collected or analyzed. Ethical approval was not required under applicable institutional policies for secondary analysis of public platform data.

## **Data availability**

Stack Overflow public data dumps were obtained from the Internet Archive (<https://archive.org/details/stackexchange>). Analysis code and step-by-step instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review). Processed analytical datasets used in the matching and survival analyses are available from the corresponding author upon reasonable request.

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## Appendix A. Replication materials

Analysis code and instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review).

## Appendix B. Matching Diagnostics

Figure B.6 shows sufficient common support between the treatment and control groups: the propensity-score distributions of the original treated and control groups are similar, and the distribution of the matched control group is nearly identical to that of the treated group.

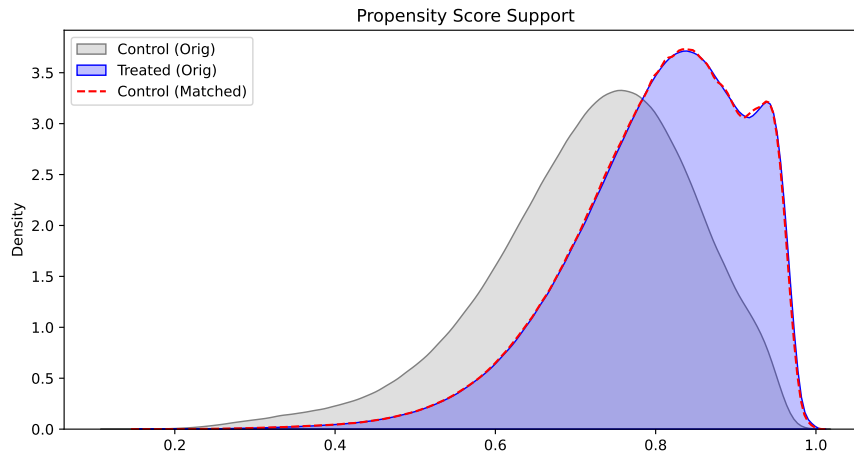


Figure B.6: Distribution of propensity scores of treatment group, control group, and matched control group

Figure B.7 displays the standardized mean differences (SMDs) for the matching covariates before and after matching. Several covariates show noticeable imbalance before matching; after matching, all SMDs are close to zero.

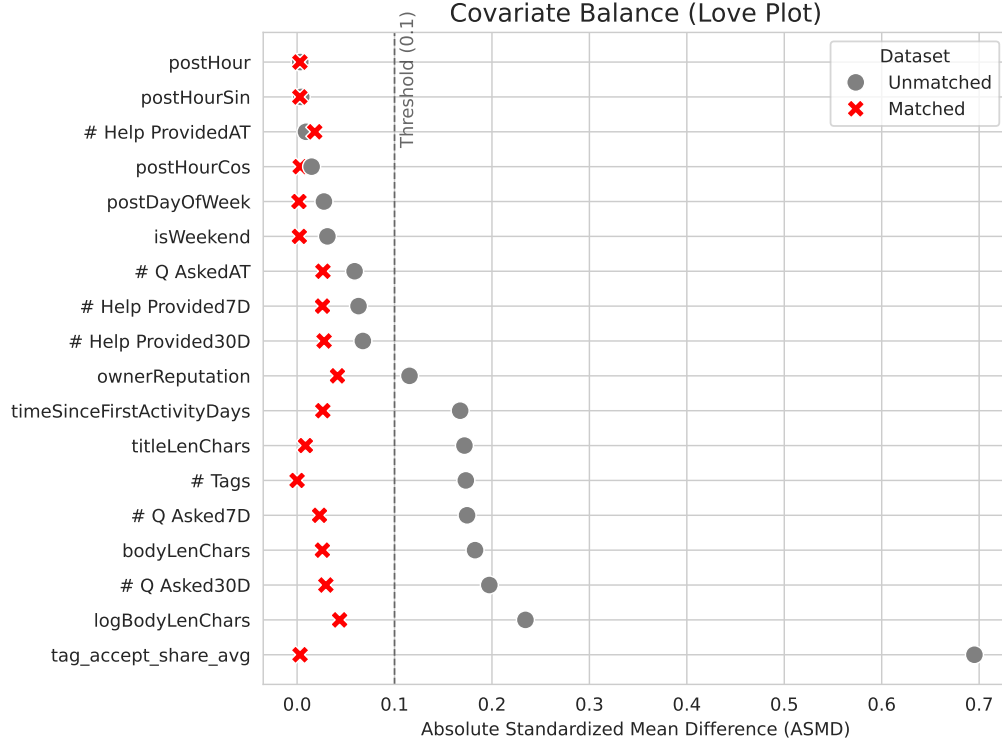


Figure B.7: Visualization of balance in the unmatched and matched dataset

## Appendix C. Common Support and the Scope of the Estimand

Matching restricts the scope of the estimand rather than the set of users we observe. Of the 4,999,330 unanswered questions in the pre-matching sample, 2,481,556 serve as matched controls. The rest either lack a valid exact stratum or complete matching covariates, or are never selected as a nearest neighbor within the caliper, so roughly half the unanswered pool has no on-support match. Because the exact strata are calendar year, primary tag, and number of tags, the control pool that survives is drawn from more readily answered tags than the discarded tail.

The reported contrast is therefore the effect for answered questions that have an on-support unanswered counterpart. This is a generalizability restriction inherent to matching, and a conservative one: because the discarded questions are the less answerable ones, the more-answerable retained controls set a higher, not lower, comparison baseline. The discarding is also not concentrated in the newcomer stratum that carries our central result, where the  $< 1$  week bucket is dropped at close to its arm-wide average rate.

## Appendix D. Help Rates Over the Observation Window by Tenure

This appendix reproduces both panels of Figure 3 separately by tenure bucket. Figure D.8 disaggregates the treated–control comparison of panel (a), and Figure D.9 the answer–receipt comparison of panel (b).

Figure D.8 shows that treated and control help rates track each other closely before the question in every tenure bucket, supporting the parallel-trends logic within each stratum. The sharp spike at question time is common to both groups and is somewhat larger for controls; the small post-answer elevations estimated in Table 7 and Figure 4 (hazard ratios of a few percent) are below what these raw normalized rates can visually resolve.

Figure D.9 shows the ordering discussed in Section 5.5 holding in every tenure stratum. Users still waiting for an answer help at the highest rate in the first half-hour after the question, and from roughly an hour onward users who have already received an answer help at a modestly higher rate than both those still waiting and the matched controls. As in the pooled panel, these are raw rates rather than estimates: the differences are small, and answer-arrival timing is not random within the treated group.

## Appendix E. Estimation Sample Sizes

Table E.11 reconciles the several sample sizes reported across the Cox tables. Each matched pair contributes one treated and one control question-level row (a control reused across pairs is counted once per pair it anchors), and the Cox partial likelihood is fit over the person-interval rows expanded from those matched rows. The pooled point estimates use the full interval set; only the matched-pair bootstrap replicates (Table F.12) cap each replicate at approximately 8 million interval rows. Users and questions are dataset-wide distinct counts and do not partition by tenure bucket, so the bucket rows report only the matched-row, interval-row, and event counts that do.

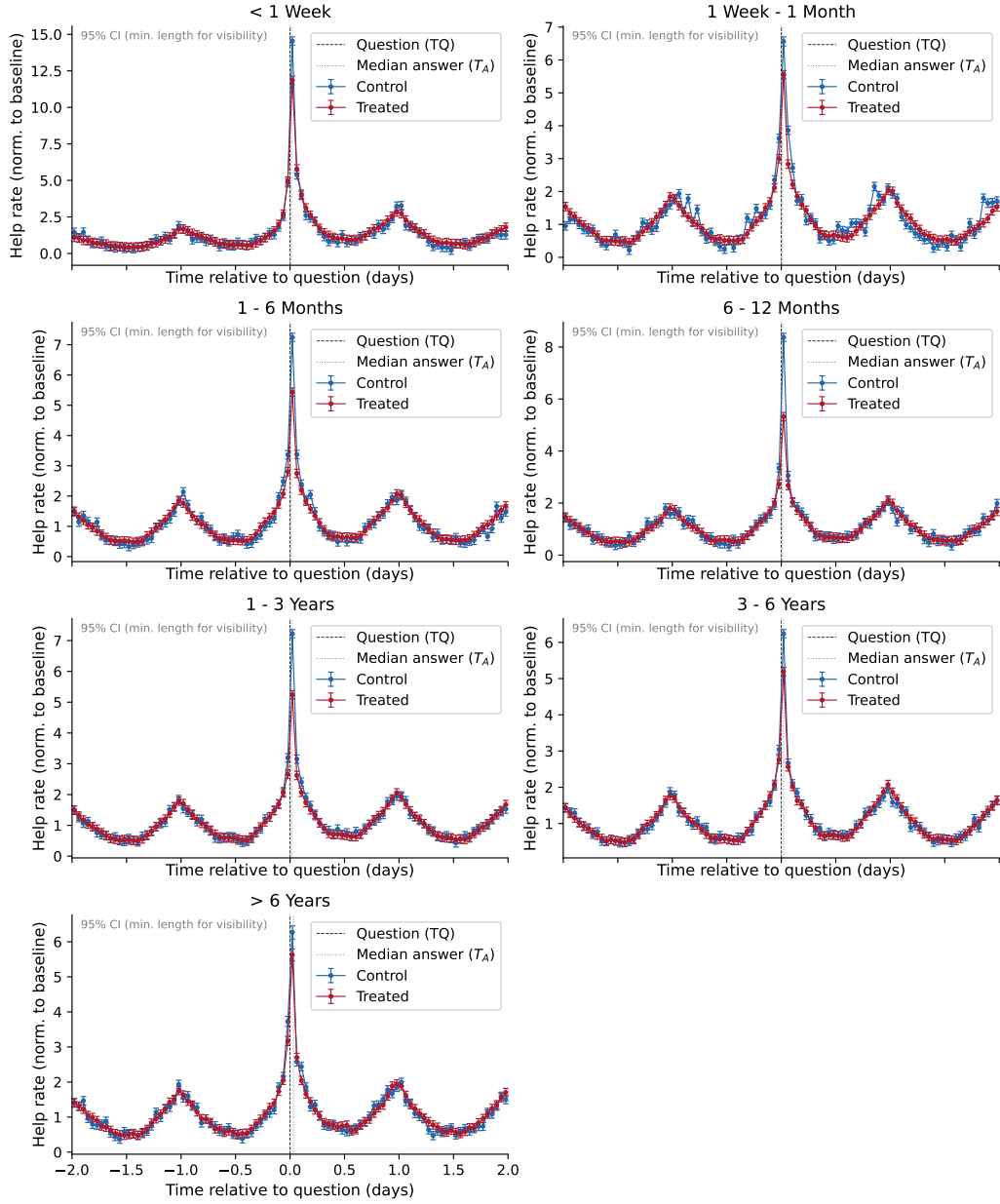


Figure D.8: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group's pre-question baseline) for treated users (red) and matched control users (blue) over the  $\pm 2$ -day window around the question-posting time ( $T_Q$ , dashed vertical line). The dotted vertical line marks the median answer arrival time ( $T_A$ ). Error bars indicate 95% confidence intervals.

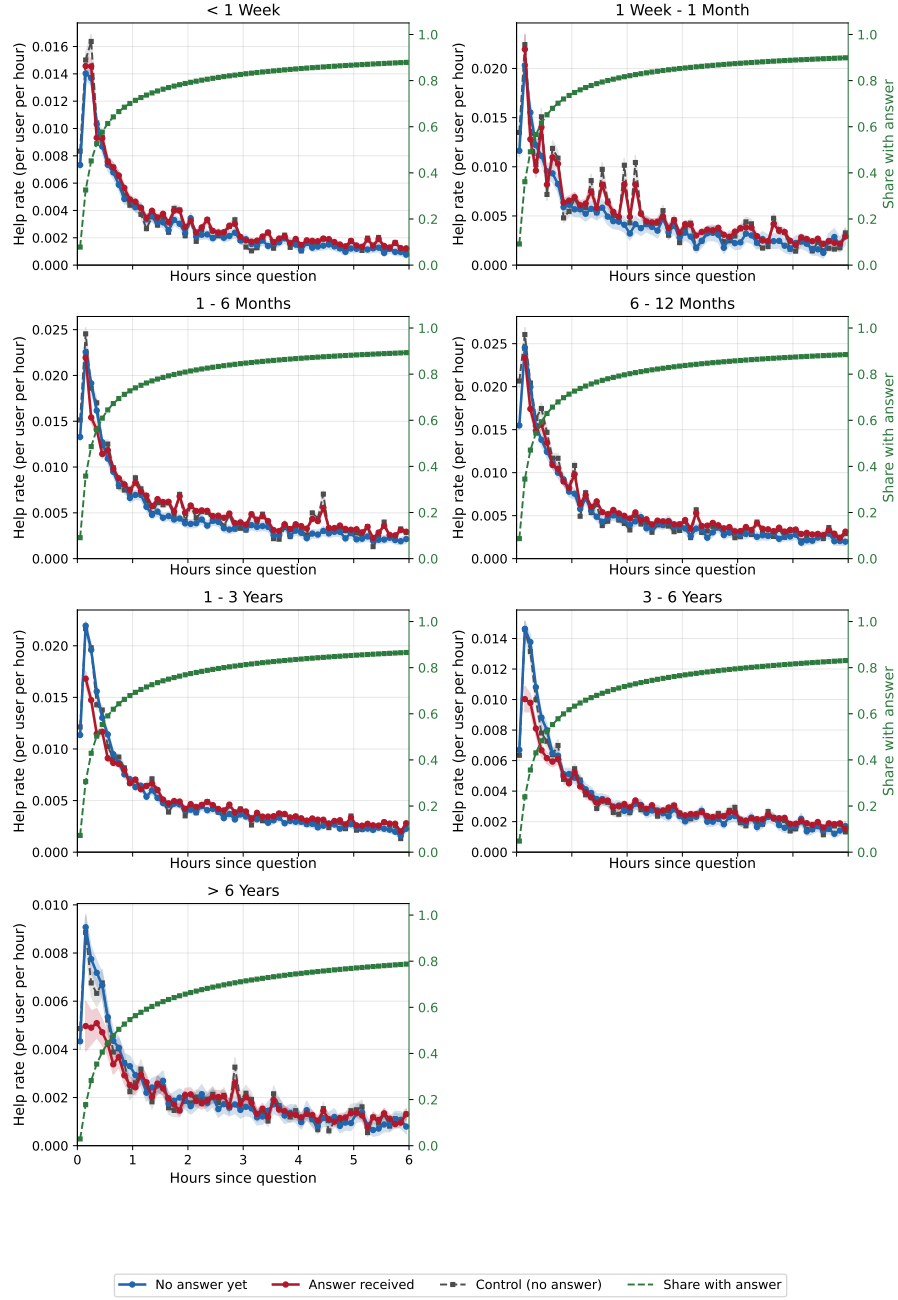


Figure D.9: Help rate in the post-question window by answer-receipt status, separately by tenure bucket. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour. This is panel (b) of Figure 3 disaggregated by tenure.

Table E.11: Estimation Sample Sizes by Model

| Model                            | Users     | Questions  | Matched rows | Interval rows | Events    |
|----------------------------------|-----------|------------|--------------|---------------|-----------|
| Pooled (Models A & B)            | 3,732,706 | 14,656,436 | 24,349,760   | 62,379,420    | 3,435,973 |
| <i>Model A, by tenure bucket</i> |           |            |              |               |           |
| < 1 Week                         | —         | —          | 4,029,008    | 10,055,987    | 368,780   |
| 1 Week - 1 Month                 | —         | —          | 1,551,586    | 3,895,000     | 238,189   |
| 1 - 6 Months                     | —         | —          | 4,045,952    | 10,248,285    | 694,001   |
| 6 - 12 Months                    | —         | —          | 2,983,180    | 7,632,877     | 537,263   |
| 1 - 3 Years                      | —         | —          | 6,424,968    | 16,588,890    | 1,037,470 |
| 3 - 6 Years                      | —         | —          | 3,614,184    | 9,444,885     | 428,036   |
| > 6 Years                        | —         | —          | 1,700,882    | 4,513,496     | 132,234   |

**Notes:** Matched rows are treated + control question-level rows in the Cox sample (=  $2 \times$  matched pairs); a control reused across pairs contributes one row per pair it anchors. Interval rows are the person-interval rows expanded from the matched rows and fit by the Cox partial likelihood; the point estimates use the full interval set, while the matched-pair bootstrap (Table F.12) caps each replicate at 8M interval rows. Users and Questions are dataset-wide distinct counts and do not partition by tenure bucket (a user’s questions, and reused control questions, can fall in several buckets), so the bucket rows leave those columns blank. Events = helping events in the full analysis sample (after time rounding).

## Appendix F. Bootstrap Inference and Absolute Effects

Table F.12 reports matched-pair bootstrap 95% confidence intervals for the Model A treatment effect — the DiD increment at answer arrival ( $\beta_4$ ) — for the pooled sample (100 replicates) and, because it carries the paper’s headline result, the newcomer (< 1 week) stratum at 500 replicates. Absolute effect sizes (absolute rate differences and the implied answers-needed-per-additional-helping-event) are reported in the main text (Table 10, “Implications for Platform Design”).

Table F.12: Matched-Pair Bootstrap Uncertainty for the Answer-Arrival DiD Increment ( $\beta_4$ ; Model A)

| Scope    | HR    | Bootstrap 95% CI | Replicates | N          |
|----------|-------|------------------|------------|------------|
| Pooled   | 1.049 | [1.032, 1.075]   | 100/100    | 24,349,760 |
| < 1 Week | 1.071 | [1.057, 1.084]   | 500/500    | 4,029,008  |

**Notes:** Replicates resample whole matched pairs with replacement. HR is the answer-arrival increment  $\exp(\beta_4)$ , the DiD treatment effect.

## Appendix G. Robustness: Cohort Heterogeneity, Observable Selection, and Answer Quality

Table G.13 addresses the concern that pooling entry cohorts from 2008 through 2025 conflates structurally different eras of the platform (reputation inflation, tag evolution, and the post-2022 generative-AI disruption). Because calendar year enters the matching design only as an exact stratum, treated and control questions within a pair always share a year; to probe heterogeneity *across* years we re-estimate Model A on the pre-2020 subsample, both pooled and in the newcomer (< 1 week) stratum. The pooled pre-2020 arrival hazard ratio is 1.04 (versus 1.05 in the full sample), and the pre-2020 newcomer estimate reproduces the full-sample figure (arrival HR = 1.07 in both), as expected since the pre-2020 rows are most of the sample, indicating that the tenure gradient was already present before the generative-AI era. Because calendar year enters as an exact-matching stratum, the pre-2020 subsample is not disjoint from the full sample but a large sub-population of it, retaining 79% of matched rows and 91% of helping events (Table G.13). This check is therefore compositional — it dates our estimates to the pre-generative-AI platform — rather than an independent replication on separate data; the disjoint post-2020 subsample is reported in the main text, where the pooled effect persists but the newcomer effect does not.

Table G.13: Cohort Heterogeneity Robustness

| Specification                    | Arrival HR | 95% CI         | Waiting $\beta_2$ | N          | Events    |
|----------------------------------|------------|----------------|-------------------|------------|-----------|
| Full sample, pooled              | 1.049      | [1.044, 1.054] | -0.007*           | 24,349,760 | 3,435,973 |
| Full sample, newcomer (< 1 week) | 1.071      | [1.056, 1.086] | +0.050***         | 4,029,008  | 368,780   |
| Pre-2020, pooled                 | 1.037      | [1.031, 1.042] | -0.000            | 19,187,320 | 3,144,155 |
| Pre-2020, newcomer (< 1 week)    | 1.068      | [1.052, 1.085] | +0.071***         | 3,155,176  | 336,966   |
| Post-2020, pooled                | 1.043      | [1.029, 1.057] | +0.011†           | 5,162,440  | 291,818   |
| Post-2020, newcomer (< 1 week)   | 1.001      | [0.968, 1.034] | -0.026            | 873,832    | 31,814    |

**Notes:** Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; † $p < .1$

Table G.14 reports the pooled Model A after adding pre-treatment observable selection controls (posting hour and weekday, number of tags, body and title length, and asker reputation). View count is *not* included: it accrues after posting and is higher for answered questions, so conditioning on it would open a post-treatment / collider channel already excluded from the propen-

sity score. The pooled arrival increment remains essentially unchanged once these pre-treatment observables are held fixed (arrival HR = 1.04, versus 1.05 in the baseline specification).

Table G.14: Observable Selection Controls

| Specification         | Arrival HR | 95% CI         | Waiting $\beta_2$ | N          | Events    |
|-----------------------|------------|----------------|-------------------|------------|-----------|
| Baseline (Model A)    | 1.049      | [1.044, 1.054] | -0.007*           | 24,349,760 | 3,435,973 |
| + observable controls | 1.040      | [1.036, 1.044] | +0.002            | 24,349,760 | 3,435,973 |

**Notes:** Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

Table G.15 reports the pooled speed-specification arrival increment as the three answer-quality controls (first-answer length, vote score, and acceptance) are entered one at a time. The gradient is informative about what each signal absorbs: answer length — the least endogenous signal, fixed by the answerer at composition — barely moves the estimate (1.038 baseline to 1.032); vote score brings it to 1.010; and the full specification, adding acceptance, pushes it below one (0.982), with the same pattern in the newcomer stratum (0.95; Table G.16). Because vote score and, especially, acceptance are post-treatment and endogenous to the outcome (accepting requires the asker to return to the site, the same re-engagement that produces helping), these level estimates are a bounding exercise rather than a causal decomposition: the association is robust to the terse-versus-detailed concern (length), and it is reversed only by the signals a bad-control logic predicts would over-adjust.

Table G.15: Answer-Quality Robustness

| Specification                   | Arrival HR | 95% CI         | Waiting $\beta_2$ | N          | Events    |
|---------------------------------|------------|----------------|-------------------|------------|-----------|
| Speed specification, baseline   | 1.038      | [1.034, 1.042] | +0.014***         | 24,349,760 | 3,435,973 |
| + answer length                 | 1.032      | [1.027, 1.036] | +0.006**          | 24,349,760 | 3,435,973 |
| + answer score                  | 1.010      | [1.006, 1.015] | -0.015***         | 24,349,760 | 3,435,973 |
| + length, score, and acceptance | 0.982      | [0.978, 0.986] | -0.046***         | 24,349,760 | 3,435,973 |

**Notes:** Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ;  $^\dagger p < .1$

Table G.16 reports the newcomer (< 1 week) bucket under several specifications. The baseline arrival increment is 1.07. Adding the same pre-treatment observable controls — still *without* view count — raises the arrival increment to 1.13, which indicates the newcomer association is not an

artifact of *observable* selection. This gain warrants caution rather than reassurance, however: the waiting-period diagnostic rises in step, from  $\beta_2 = +0.050$  to  $+0.122$ , essentially the magnitude of the arrival log-increment itself ( $\log 1.13 = 0.121$ ). Under these controls, treated newcomers already diverge upward from their matched controls by nearly the full effect size *before* the answer arrives, so the two terms inflate together. We therefore read the specification as ruling out observable selection as the sole driver, not as independent corroboration of the arrival increment; it also sharpens the parallel-trends caveat that already attaches to this stratum. Adding answer-quality controls, by contrast, pushes the newcomer increment below one (arrival 0.95), but this mirrors the pooled pattern and, as above, reflects over-adjustment on post-treatment signals (vote score and acceptance) rather than a quality confound the design fails to resolve.

Table G.16: Newcomer Bucket Robustness (< 1 Week)

| Specification                   | Arrival HR | 95% CI         | Waiting $\beta_2$ | N         | Events  |
|---------------------------------|------------|----------------|-------------------|-----------|---------|
| Model A, baseline               | 1.071      | [1.056, 1.086] | +0.050***         | 4,029,008 | 368,780 |
| Model A + observable controls   | 1.129      | [1.117, 1.142] | +0.122***         | 4,029,008 | 368,780 |
| Model B (speed), baseline       | 1.010      | [0.999, 1.021] | +0.008            | 4,029,008 | 368,780 |
| Model B + answer length         | 1.003      | [0.992, 1.014] | -0.001            | 4,029,008 | 368,780 |
| Model B + answer score          | 0.979      | [0.968, 0.990] | -0.022***         | 4,029,008 | 368,780 |
| Model B + full quality controls | 0.950      | [0.939, 0.960] | -0.052***         | 4,029,008 | 368,780 |

**Notes:** Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ;  $^\dagger p < .1$

Table G.17 revisits the discrete response-time bins after adding the same three answer-quality controls used in the pooled speed specification (acceptance, first-answer score, and length). Because each row already conditions on a response-time window, the quality-controlled fits use Model A covariates plus those three signals rather than continuous log-response-time interactions. Comparing columns shows how much of the bin-wise arrival pattern survives once observable quality is held fixed.

Table G.17: Treatment Effect by Response-Time Bin, With and Without Answer-Quality Controls

| Response time | Baseline HR | 95% CI         | Quality HR | 95% CI         | N          |
|---------------|-------------|----------------|------------|----------------|------------|
| 0-15 min      | 0.998       | [0.989, 1.007] | 0.953***   | [0.947, 0.959] | 17,412,740 |
| 15-30 min     | 1.004       | [0.989, 1.019] | 0.951***   | [0.941, 0.962] | 13,934,456 |
| 30-60 min     | 1.190***    | [1.174, 1.206] | 1.029***   | [1.018, 1.040] | 13,549,736 |
| 1-2 hr        | 1.056***    | [1.040, 1.072] | 0.965***   | [0.953, 0.976] | 13,211,144 |
| 2-4 hr        | 1.030***    | [1.013, 1.047] | 0.961***   | [0.948, 0.974] | 12,958,927 |
| 4-8 hr        | 1.017†      | [0.999, 1.035] | 0.968***   | [0.953, 0.982] | 12,762,008 |
| 8-12 hr       | 1.019       | [0.995, 1.044] | 0.971**    | [0.951, 0.991] | 12,464,162 |
| 12-24 hr      | 1.048***    | [1.028, 1.067] | 1.008      | [0.993, 1.024] | 12,645,932 |
| 1-3 days      | 1.088***    | [1.069, 1.108] | 1.089***   | [1.071, 1.106] | 12,592,667 |
| >3 days       | 1.015       | [0.992, 1.038] | 1.041***   | [1.020, 1.063] | 12,391,574 |

**Notes:** Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool. Bins are cut on the actual question-to-answer latency ( $T_A - T_Q$ ); because the observation window closes two days after the answer, every treated question retains a full two-day post-answer phase, so  $\beta_4$  is identified in all bins including >3 days. HR is the answer-arrival increment ( $\beta_4$ ). Quality columns add acceptance, first-answer score, and length.

Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; † $p < .1$

Table G.18 probes whether the answer-quality result depends on how the vote-score control is coded. The specifications in Table G.15 enter the score linearly (winsorized at the 5th and 95th percentiles and standardized); here we re-estimate the score-only and full-quality specifications, pooled and in the newcomer stratum, under two alternatives — a signed-log (asinh) coding and ordinal bins (1–2, 3–9, and  $\geq 10$  votes, reference  $\leq 0$ ) — at full data. We use asinh(score) rather than  $\log(1 + \text{score})$  because roughly two million answers carry net-negative (downvoted) scores, for which  $\log(1 + \text{score})$  is undefined. These negative-scored cases are treated questions whose earliest reply within the window was a downvoted answer that arrived before the later non-negative answer qualifying the question as treated; because *firstAnswerScore* and the response-time clock  $T_{\text{answer}}$  are keyed to that first reply rather than to the qualifying one, both can reflect the downvoted answer rather than the one that earns treatment status. The below-one full-specification result is robust to the coding: the pooled arrival increment is 0.98 under all three codings and the newcomer counterpart 0.95, and if anything the better-specified signed-log and binned codings pull the estimate marginally *further* below one rather than back toward it. The one estimate that does not

survive re-coding is the intermediate score-only step, which is significantly above one under the linear coding (1.01) but statistically indistinguishable from one under the signed-log and binned codings; we therefore do not read it as evidence of an above-one effect.

Table G.18: Answer-Score Coding Sensitivity

| Specification                 | Score coding | Arrival HR | 95% CI         | Waiting $\beta_2$ | N          | Events    |
|-------------------------------|--------------|------------|----------------|-------------------|------------|-----------|
| <i>Pooled</i>                 |              |            |                |                   |            |           |
| Speed baseline                | —            | 1.038***   | [1.034, 1.042] | +0.014***         | 24,349,760 | 3,435,973 |
| + Vote score                  | Linear       | 1.010***   | [1.006, 1.015] | -0.015***         | 24,349,760 | 3,435,973 |
|                               | Asinh        | 1.001      | [0.997, 1.005] | -0.026***         | 24,349,760 | 3,435,973 |
|                               | Ordinal bins | 1.003      | [0.999, 1.007] | -0.023***         | 24,349,760 | 3,435,973 |
|                               | Linear       | 0.982***   | [0.978, 0.986] | -0.046***         | 24,349,760 | 3,435,973 |
| + Full quality controls       | Asinh        | 0.977***   | [0.973, 0.982] | -0.052***         | 24,349,760 | 3,435,973 |
|                               | Ordinal bins | 0.979***   | [0.975, 0.983] | -0.050***         | 24,349,760 | 3,435,973 |
| <i>Newcomer (&lt; 1 Week)</i> |              |            |                |                   |            |           |
| Speed baseline                | —            | 1.010†     | [0.999, 1.021] | +0.008            | 4,029,008  | 368,780   |
| + Vote score                  | Linear       | 0.979***   | [0.968, 0.990] | -0.022***         | 4,029,008  | 368,780   |
|                               | Asinh        | 0.974***   | [0.963, 0.985] | -0.028***         | 4,029,008  | 368,780   |
|                               | Ordinal bins | 0.976***   | [0.965, 0.987] | -0.026***         | 4,029,008  | 368,780   |
|                               | Linear       | 0.950***   | [0.939, 0.960] | -0.052***         | 4,029,008  | 368,780   |
| + Full quality controls       | Asinh        | 0.947***   | [0.936, 0.957] | -0.056***         | 4,029,008  | 368,780   |
|                               | Ordinal bins | 0.949***   | [0.938, 0.959] | -0.054***         | 4,029,008  | 368,780   |

**Notes:** Codings of the answer’s vote score, all re-expressing the same control: *Linear* is the control tabled in Table G.15 (raw *Score* winsorized at the 5th/95th percentiles and standardized); *Asinh* is the inverse hyperbolic sine  $\text{asinh}(\text{score}) = \ln(\text{score} + \sqrt{\text{score}^2 + 1})$  (a signed log, concave in the score and defined for the negative scores present), likewise winsorized and standardized; *Ordinal bins* are indicators for 1–2, 3–9, and  $\geq 10$  votes (reference score  $\leq 0$ ). Arrival-HR stars test  $H_0$ : HR = 1. The bin reference is score  $\leq 0$  and the concave coding uses asinh rather than  $\log(1 + \text{score})$  because roughly two million answers carry net-negative (downvoted) scores, for which  $\log(1 + \text{score})$  is undefined. Estimand ( $\beta_4/\beta_2$ ), standard-error, and Events conventions follow Table 7.

\*\*\* $p < .001$ ; \*\* $p < .01$ ; \* $p < .05$ ; † $p < .1$

## Appendix H. Implementation Details

All models are estimated using the `lifelines CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is  $\lambda = 5 \times 10^{-3}$  for main-effect models and  $\lambda = 10^{-2}$  for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk

sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. Before each fit, interval boundaries are rounded to a one-hour grid (interval start and stop times to the nearest hour) and zero-length intervals produced by the rounding are dropped. Multiple helping events by the same user within the same hour of a window therefore collapse to a single counted event. This is the “time rounding” referenced in the table notes; all reported *Events* counts are computed after it. It is also why two families of event counts differ by construction. The raw mean of 0.18 help events per window (Table 4) corresponds to 0.1411 counted events per row in the estimation sample (3,435,973/24,349,760), and the composite-outcome Events (Table 6) are smaller than the sum of their component channels, because same-hour events across channels likewise merge. Cox-table standard errors are model-based (inverse-Hessian) and are *not* cluster-robust; dependence from matched pairs, reused controls, and repeated intervals can make those intervals too narrow. For the Model A DiD we therefore also report matched-pair bootstrap 95% CIs (resampling whole `match_id` pairs) in Table F.12 — 100 replicates for the pooled effect and 500 for the newcomer (< 1 week) stratum that carries the headline result. For the pooled models (H1, spanning all experience levels), the reported point estimates are fit on the full interval frame (the fit cap is raised for the point-estimate fits), so no subsampling is applied; the default cap of approximately 8 million matched-pair user-interval rows (preserving pair integrity) applies only to the pooled bootstrap replicates, whereas the smaller newcomer stratum is fit in full within each replicate. A single 8 million-row subsample proved seed-fragile for the pooled arrival increment, which is why the reported estimate comes from a full-data fit. Table *Events* counts refer to the full analysis sample throughout, and the estimation-sample sizes are tabulated in Table E.11. Tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.